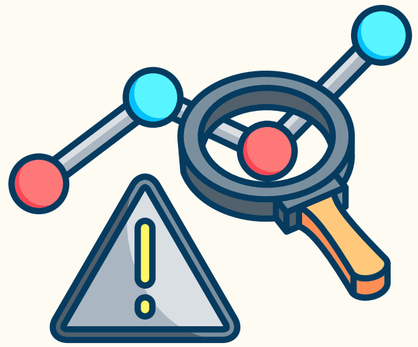


ANOMALY DETECTION IN INDUSTRIAL CONTROL SYSTEMS

STUDY PROJECT

GROUP 6: ALEKSA & SWETHA

February 05, 2025

WHAT IS AN INDUSTRIAL CONTROL SYSTEM (ICS)?

- An Industrial Control System (ICS) is a cyber-physical system that monitors and controls critical infrastructures.
- Integrates IT (Information Technology) & OT (Operational Technology).
- Ensures real-time monitoring & automation.
- Highly vulnerable to cyber-physical threats.

Source:https://www.alltialloys.com/blog-posts/permalloy-the-high-performance-alloy

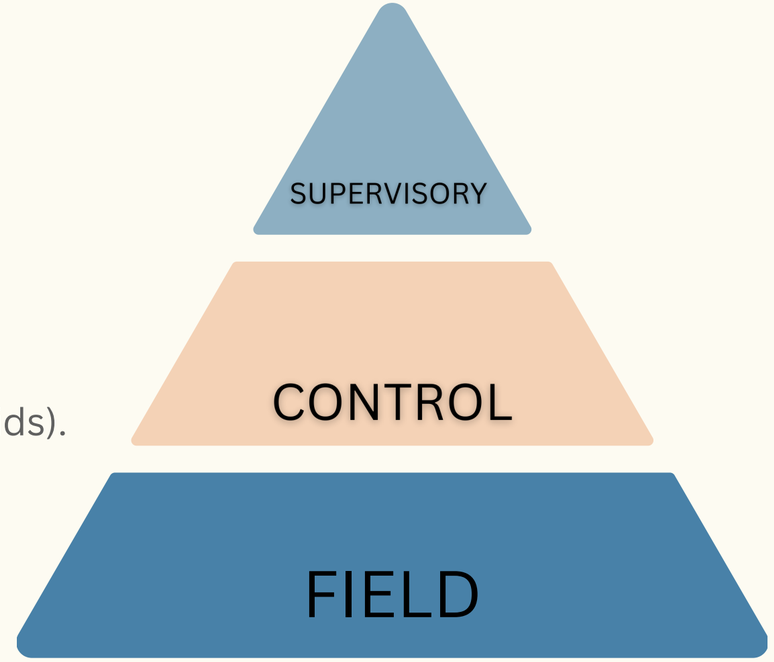
LAYERS - INDUSTRIAL CONTROL SYSTEM

3 layers:

1️⃣ Field Layer (Sensors & Actuators)
- Sensors (e.g., temperature, pressure) collect physical readings.
- Actuators (e.g., motors, valves) control physical processes.
- Communicates with controllers using industrial protocols (e.g., Modbus).

2️⃣ Control Layer (PLCs & RTUs)
- PLCs (Programmable Logic Controllers) process sensor data & make decisions.
- RTUs (Remote Terminal Units) manage remote operations (e.g., pipelines, power grids).
- Sends control signals to actuators based on predefined logic.

3️⃣ Supervisory Layer (SCADA & HMI)
- SCADA (Supervisory Control & Data Acquisition) monitors & controls industrial processes.
- HMI (Human-Machine Interface) provides a visual dashboard for operators.
- Collects & stores data for analysis, alarms, & remote management.



PROJECT GOAL

🔹 Develop and evaluate advanced anomaly detection methods for Industrial Control Systems (ICS) using statistical, traditional machine learning, and deep learning approaches.

Why is This Important?

⚡ ICS Security is Critical: Power grids, water distribution, and gas pipelines rely on ICS for safe and efficient operation.

🌍 Increased Connectivity = Higher Risk: The integration of Operational Technology (OT) and IT introduces new cyber-physical attack surfaces.

🔍 Limitations of Traditional IDS: Existing Intrusion Detection Systems (IDS) struggle to differentiate between normal variations and real threats, leading to false alarms or undetected attacks.

METHODOLOGY

1 DATASET
- WDT & WADI 2019
- Similarity between datasets
- Parsing network data & comparison
- Statistical analysis - physical readings

2 NETWORK DATASET GENERATION
- CIP - WADI

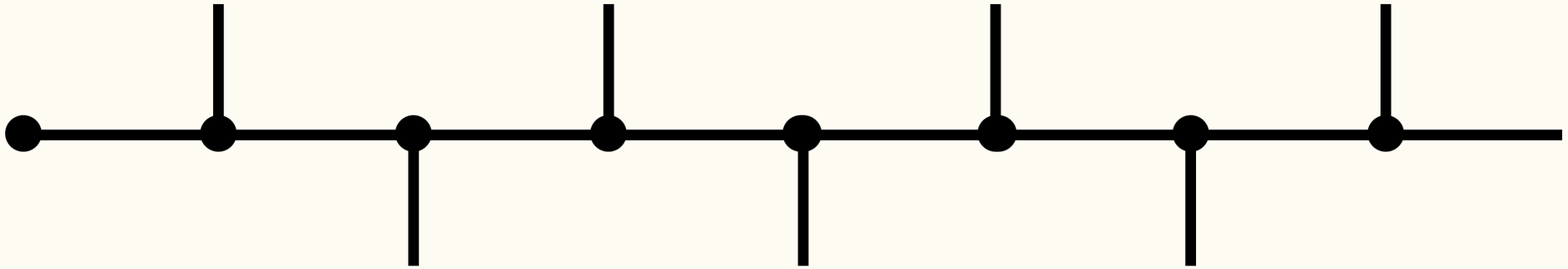
3 STATISTICS-BASED DETECTION
- Cumulative Sum(Cusum)
- Exponentially Weighted Moving Average(EWMA)

4 FEATURE EXTRACTION
- Principal Component Analysis(PCA)
- N-Gram.
- Autoencoder.

5 TRADITIONAL MACHINE LEARNING-BASED DETECTION
- k-fold cross-validation
- Train classifiers - extracted features.

6 DEEP LEARNING-BASED DETECTION & ENSEMBLE CLASSIFIER
- Convolutional Neural Network(CNN)
- One-Class and Multi-Class models

7 EVALUATION
- Scenarios
- Results



DATASETS

	WDT	WADI 2019
Source	Acquired from a water distribution testbed	Designed by iTrust, SUTD for ICS security research
Infrastructure	8 tanks, 22 solenoid valves, 6 pumps, 8 pressure sensors, 4 flow sensors, and 8 manual valves	123 sensors and actuators
Testbed Components	• Physical partition (real hardware) • Simulated partition (virtual connection)	Fully operational water distribution system
Duration	4 sessions (~2 hours total)	16 days total • 14 days normal • 2 days attack scenarios
Size	Physical dataset ~ 4.4 MB Network dataset ~10.4 GB	Physical dataset ~575 MB

SIMILARITY BETWEEN DATASETS

- Analyze WDT and WADI 2019 datasets to compare physical readings.
- Compute and visualize Euclidean distance between time steps to detect variations.
- Separate computation for:
    - Normal behavior only
    - Normal + malicious behavior.

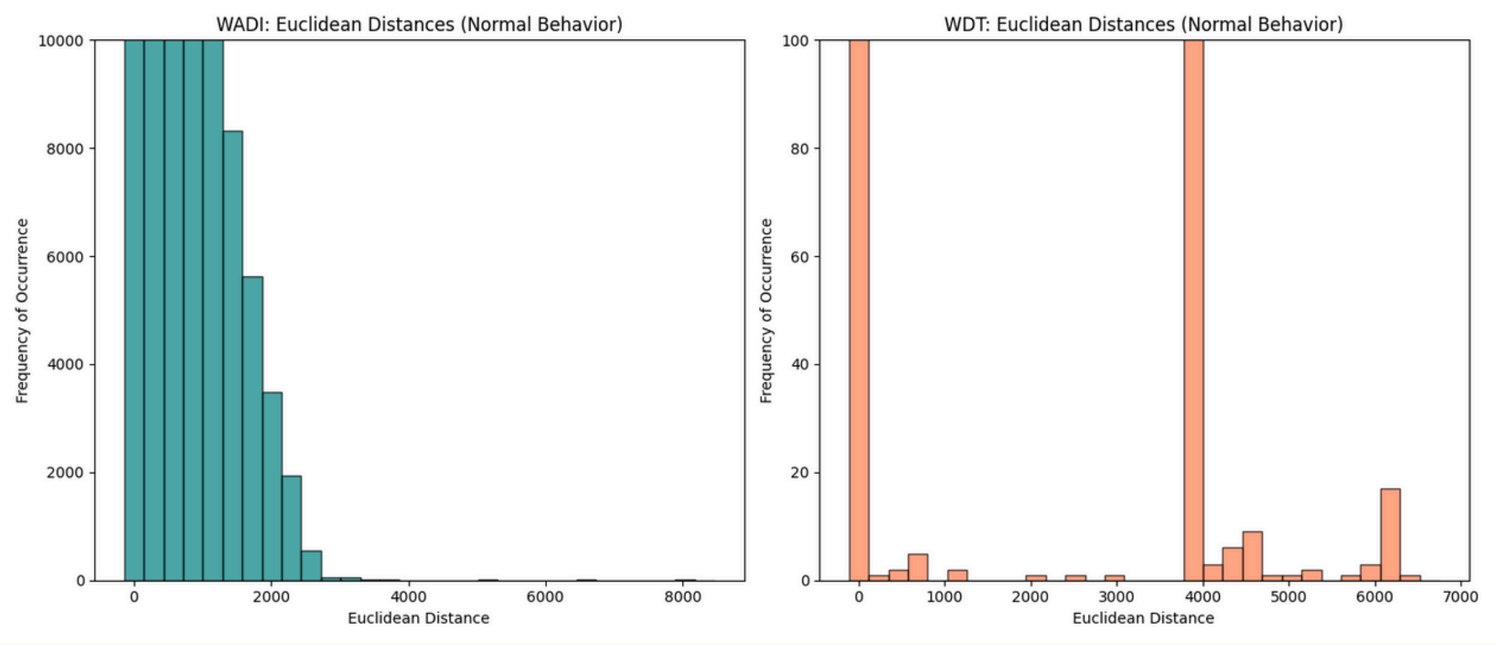
WHAT DOES THIS ACHIEVE?

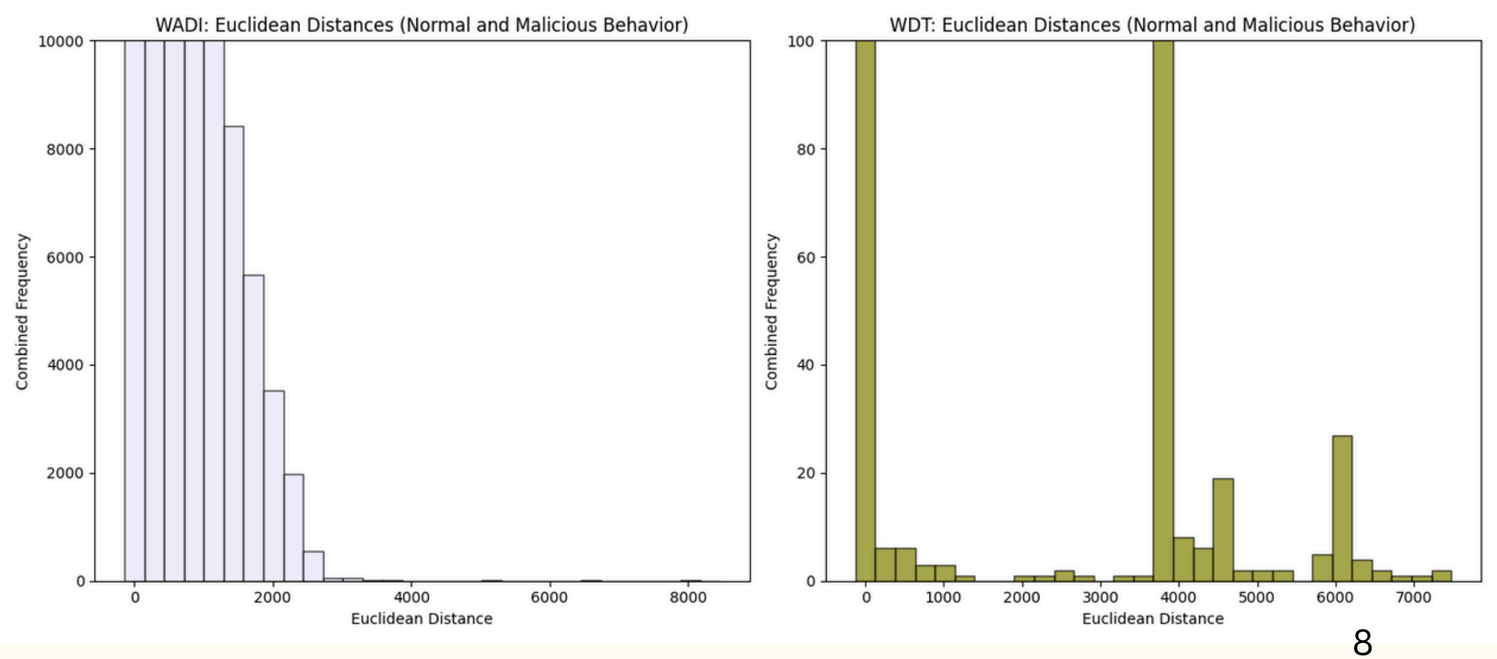
- Compare datasets(WDT and WADI 2019) to see how similar they are.
- Understanding dataset differences ensures that machine learning models generalize well.

SIMILARITY BETWEEN DATASETS

- This histogram visualizes the contrast between normal and anomalous data, which is important for feature selection and model performance.

- Large Euclidean distances in the combined behavior plot indicate attacks or anomalies, whereas smaller distances in the normal behavior plot suggest stable, expected behavior.





PARSING NETWORK PACKETS & DATA COMPARISON

- Use Scapy to process raw pcap files.
- Convert extracted data into a human-readable format.

Key Steps:

- Parse Modbus TCP packets to retrieve sensor/actuator readings.
- Store parsed data in a structured table (rows: time periods, columns: sensors/actuators).
- Compute MinMax statistics for parsed readings:
  - Calculate min/max values from original dataset.
  - Introduce error margin to define range.
  - Identify deviations in newly parsed data.

$$\text{minerr} = \text{min} - (\text{max} - \text{min})/2$$
$$\text{maxerr} = \text{max} + (\text{max} - \text{min})/2$$

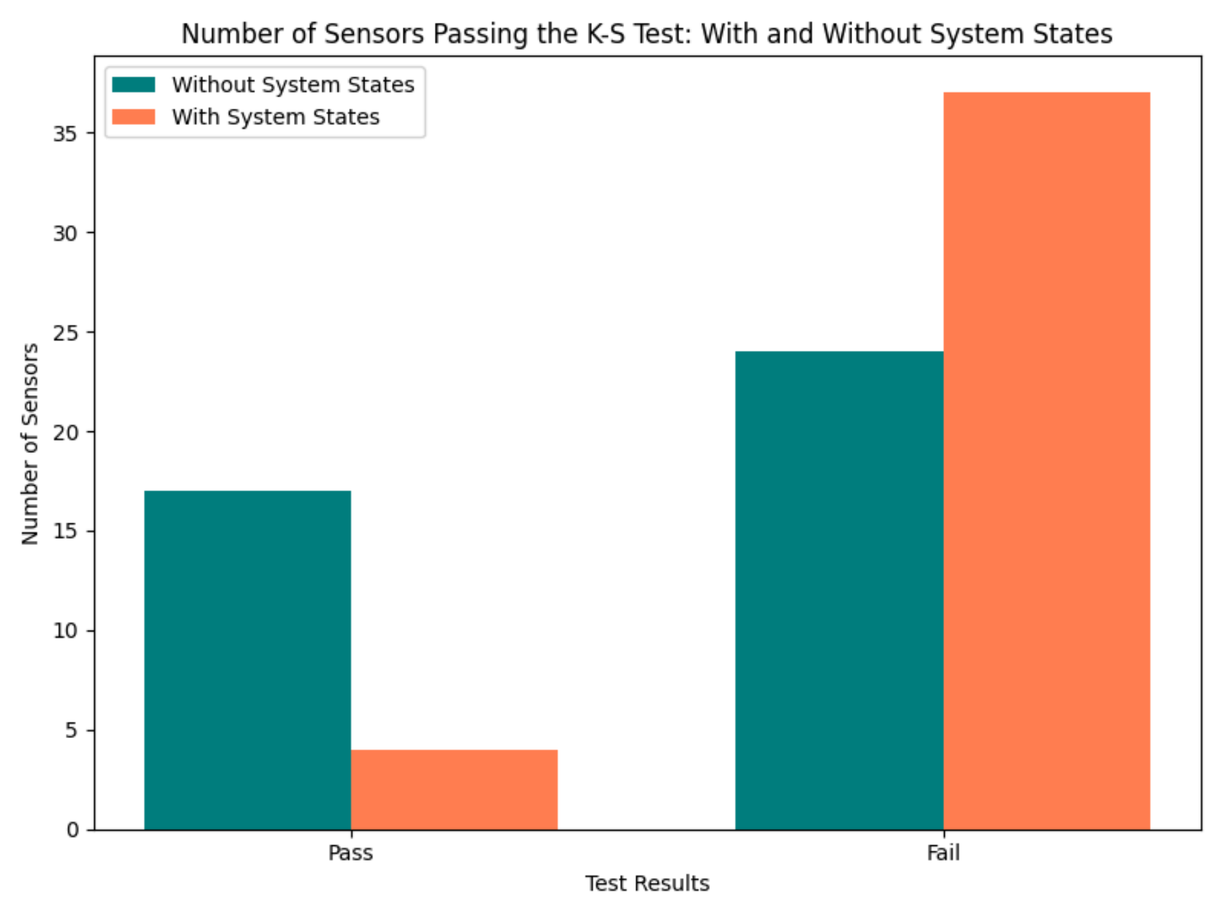
STATISTICAL ANALYSIS - PHYSICAL READINGS

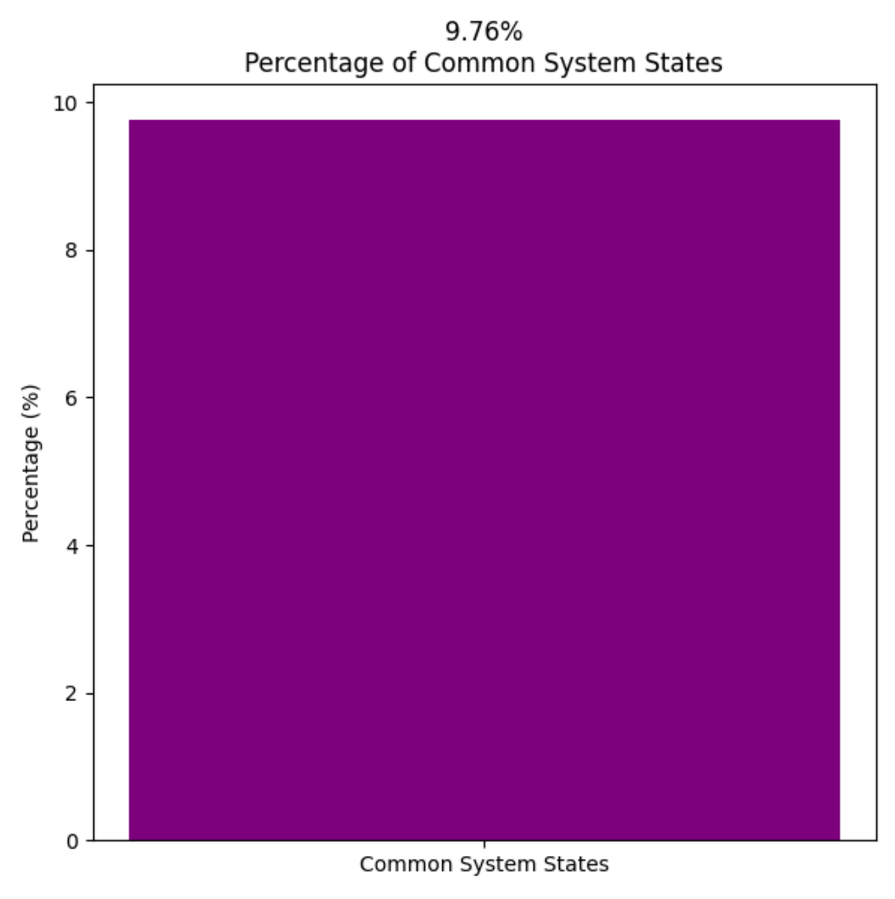
- Apply statistical techniques to detect stable patterns in normal sensor readings.
- Use the Kolmogorov-Smirnov (K-S) test to check if training and testing sets follow similar distributions.
- Identify common system states (sensor & actuator values occurring together).

WHY IS THIS IMPORTANT?

- Machine learning models work best when train and test data follow the same statistical distribution.
- If training data doesn't match real-world testing data, the model will fail to detect anomalies properly.
- Identifying system states helps in reducing false alarms by considering contextual dependencies.

STATISTICAL ANALYSIS - PHYSICAL READINGS





- It assesses the statistical similarity between training and testing data distributions.
- 9% common system state suggests that there is only a small overlap of system states between training and testing sets.

NETWORK DATASET GENERATION

🔹Convert sensor readings into network traffic using CIP(Common Industrial Protocol) in WADI dataset.
🔹Simulate realistic ICS network data for intrusion detection.

WHY GENERATE NETWORK DATASET?
✅ Needs Both Network & Sensor-Based Detection -> comparison for future experiments
✅ Testing Intrusion Detection on Realistic Network Data

Function codes	Command	Protocol
0x4C	Read Data	CIP
0x4D	Write Data	CIP

STATISTICS-BASED DETECTION

FEATURE	CUSUM(Cumulative Sum)	EWMA(Exponentially Weighted Moving Average)
Formula	**$S_0=0$, $S_{k+1} = \max(0, S_k + \lvert r_k \rvert - ß)$** r_k : current measurement - previous measurement β: threshold	**$S_0 = y_0$, $S_{k+1} = \alpha \cdot y_{k+1} + (1 - \alpha) \cdot S_k$** y_k: Current sensor reading. α: Weighting factor.
Deviation	Slow, Gradual anomalies	Sudden changes
Sensitivity	Small deviations over time	Short-term fluctuations
Computational Cost	Low (suitable for real-time)	Low (suitable for real-time)
Example	A slow modification of sensor readings (e.g., an attacker gradually increases temperature values to avoid detection).	A sudden sensor spike (e.g., a pressure sensor jumps from 20 to 80 instantly due to a cyber-attack manipulating Modbus packets).

- Optimization of parameters α and ß - counting the amount of false alarms

FEATURE EXTRACTION

1. FEATURE EXTRACTION - PCA (PRINCIPAL COMPONENT ANALYSIS)

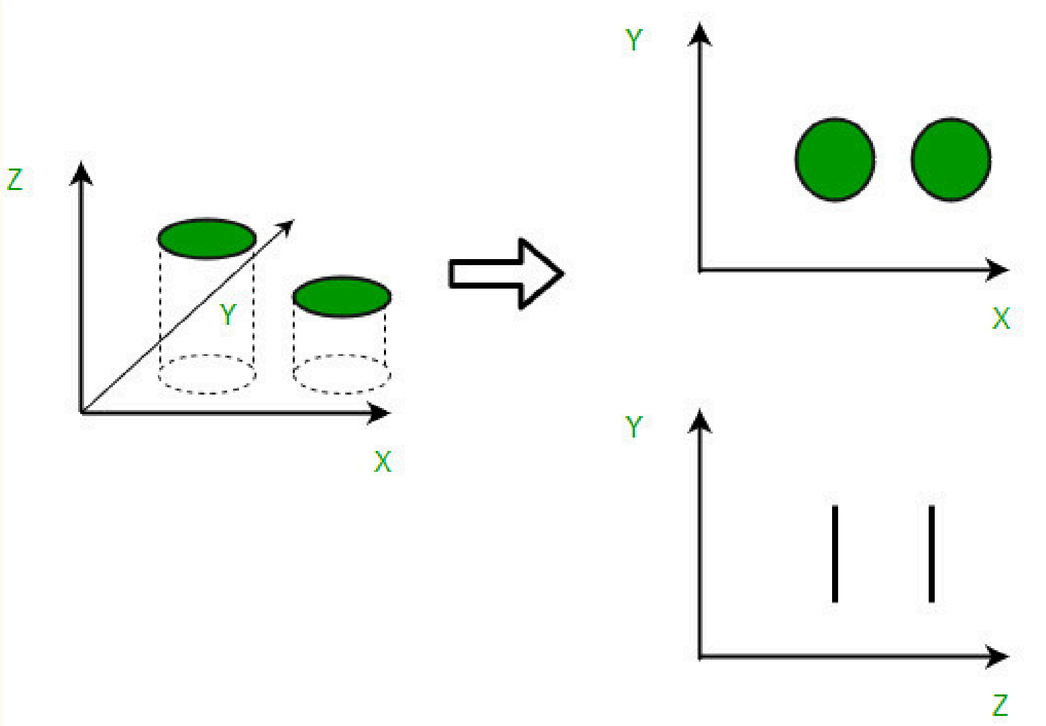
- Reduce dataset dimensionality

WORKING?

1️⃣ Computes the covariance matrix to understand relationships between features.

2️⃣ Finds eigenvalues & eigenvectors to identify the most significant components.

3️⃣ Selects top principal components that explain most of the variance.

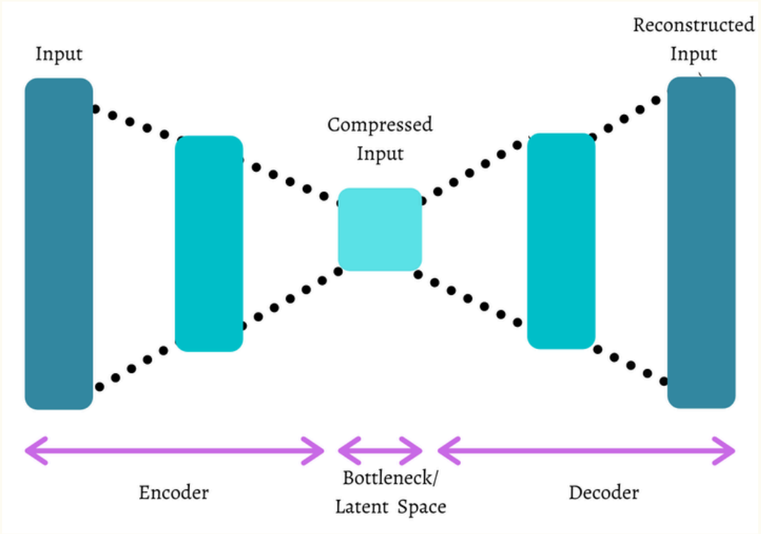


Source: https://beyondknowledgeinnovation.ai/unsupervised-learning/dimensionality-reduction/

FEATURE EXTRACTION

2. FEATURE GENERATION - AUTOENCODER

- A deep learning model that learns compressed representations of data.
- It consists of:
  - Encoder: Compresses input into a lower-dimensional representation.
  - Decoder: Reconstructs input from the compressed representation.



How Does It Work?

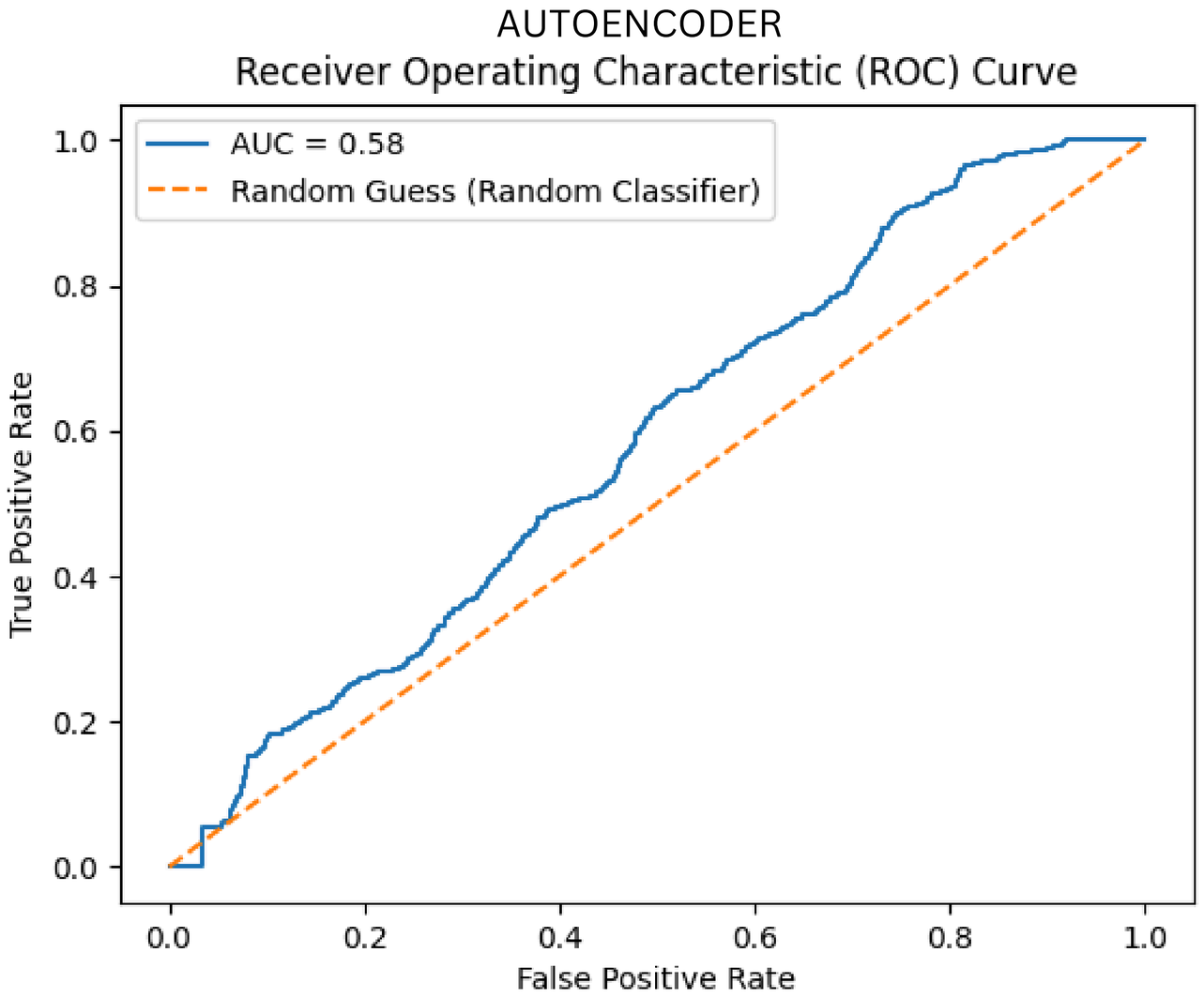
1️⃣ Train the autoencoder on data to learn important features.
2️⃣ Use the encoded layer as the extracted feature representation.
3️⃣ High reconstruction error = Potential anomaly detected.

Why is Autoencoder Important?

✅ Learns complex feature relationships that traditional methods miss.
✅ Can be trained unsupervised (no attack labels required).
✅ Useful for detecting unseen anomalies in ICS data.

Source: https://towardsdatascience.com/autoencoders-in-practice-dimensionality-reduction-and-image-denoising-ed9b9201e7e1

FEATURE EXTRACTION



- A distribution plot showing reconstruction errors for normal and anomalous data.
- The curve shows that the autoencoder is identifying attacks without classifying too many normal instances as attacks

FEATURE EXTRACTION

3. FEATURE EXTRACTION - N-GRAMS

- An N-Gram is a sequence of N values.
- Feature sets: Generate N-grams
  - Without smoothing: From raw data sequences.
  - With smoothing: Apply a smoothing technique

HOW DOES IT WORK?

1️⃣ Convert time-series sensor readings into overlapping sequences.
2️⃣ Extract patterns of normal behavior using historical data.
3️⃣ Identify anomalies when new patterns do not match expected sequences.

WHY IS N-GRAM IMPORTANT?

✅ Works well for detecting recurring attack patterns.
✅ Can be applied to both sensor & network data.

TRADITIONAL ML-BASED DETECTION

K- FOLD CROSS-VALIDATION

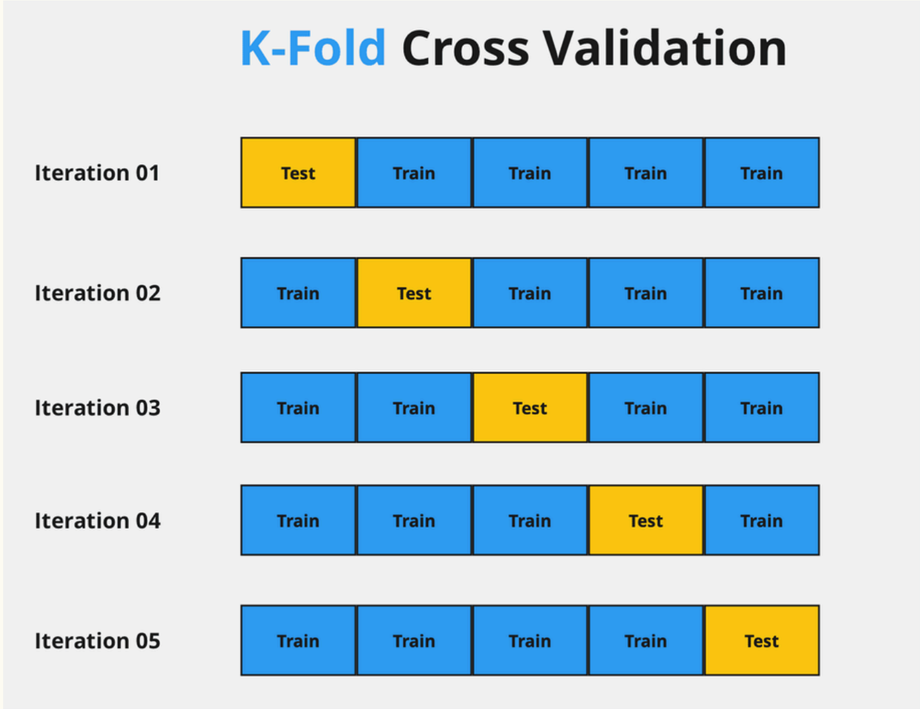
- A technique used to split the dataset into multiple training and testing sets to improve model evaluation.

HOW DOES IT WORK?

1️⃣ Divide the dataset into K equal parts (**Folds**).
2️⃣ Train the model on K-1 folds and test on the remaining fold.
3️⃣ Repeat K times, using a different fold as the test set each time.
4️⃣ Compute the average performance across all K runs.

WHY K-FOLD?

✅ Reduces bias by training on different parts of the dataset.
✅ Prevents overfitting to a single dataset split.



Source: https://dataaspirant.com/cross-validation/

SCENARIOS

SCENARIO

1

Train Data: Only normal sensor readings (no attacks).

Test Data: Contains normal readings + all attack types (each attack type equally represented).

2

Train Data: Normal data + n−1 attack types.

Test Data: Normal data + all attack types.

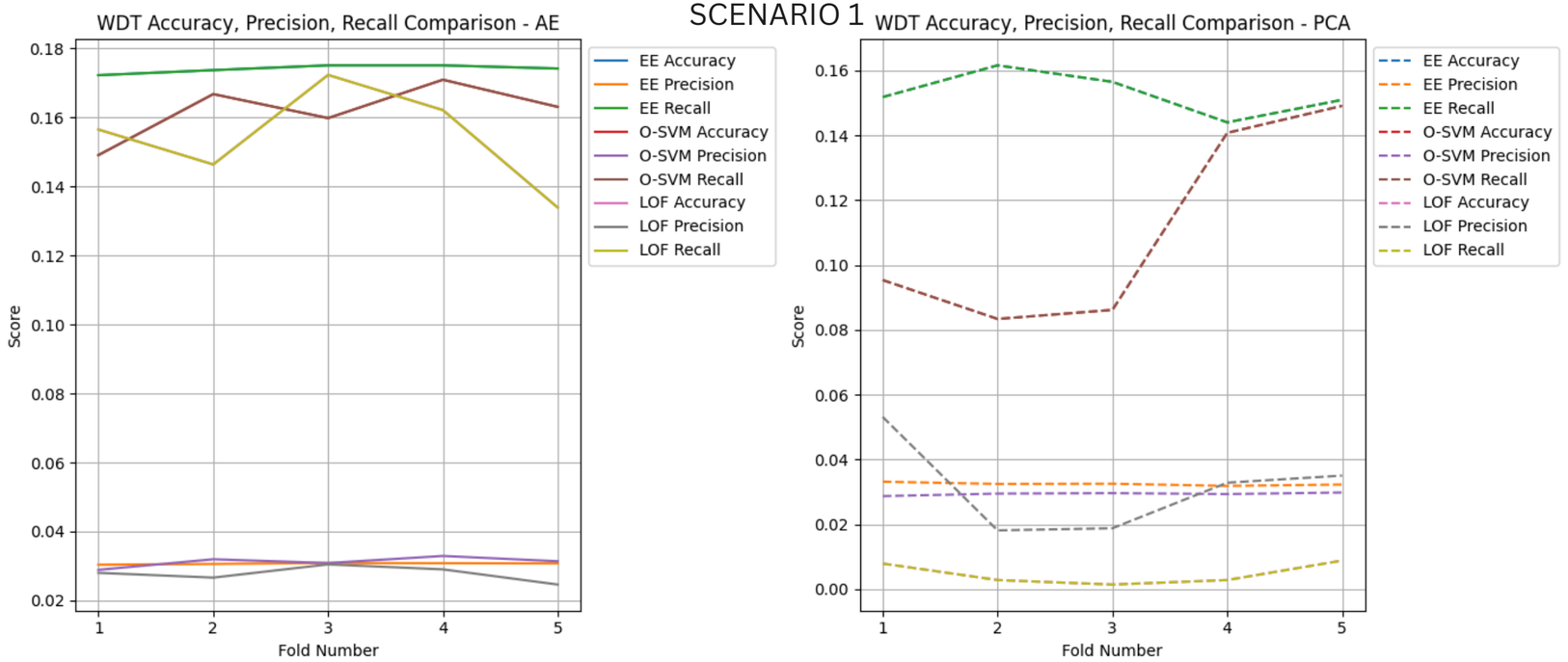
3

Train Data: Normal data + only one attack type.

Test Data: Normal data + all attack types

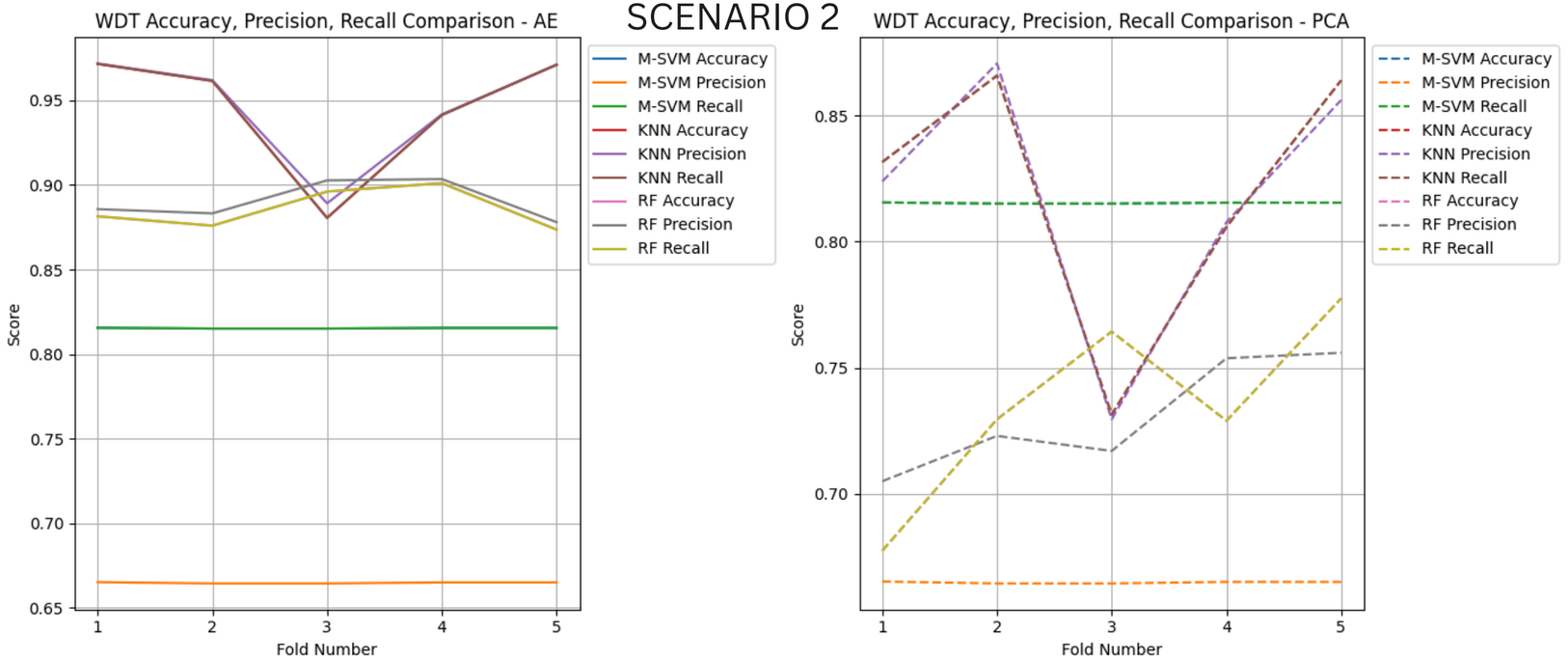
RESULTS

SCENARIO 1

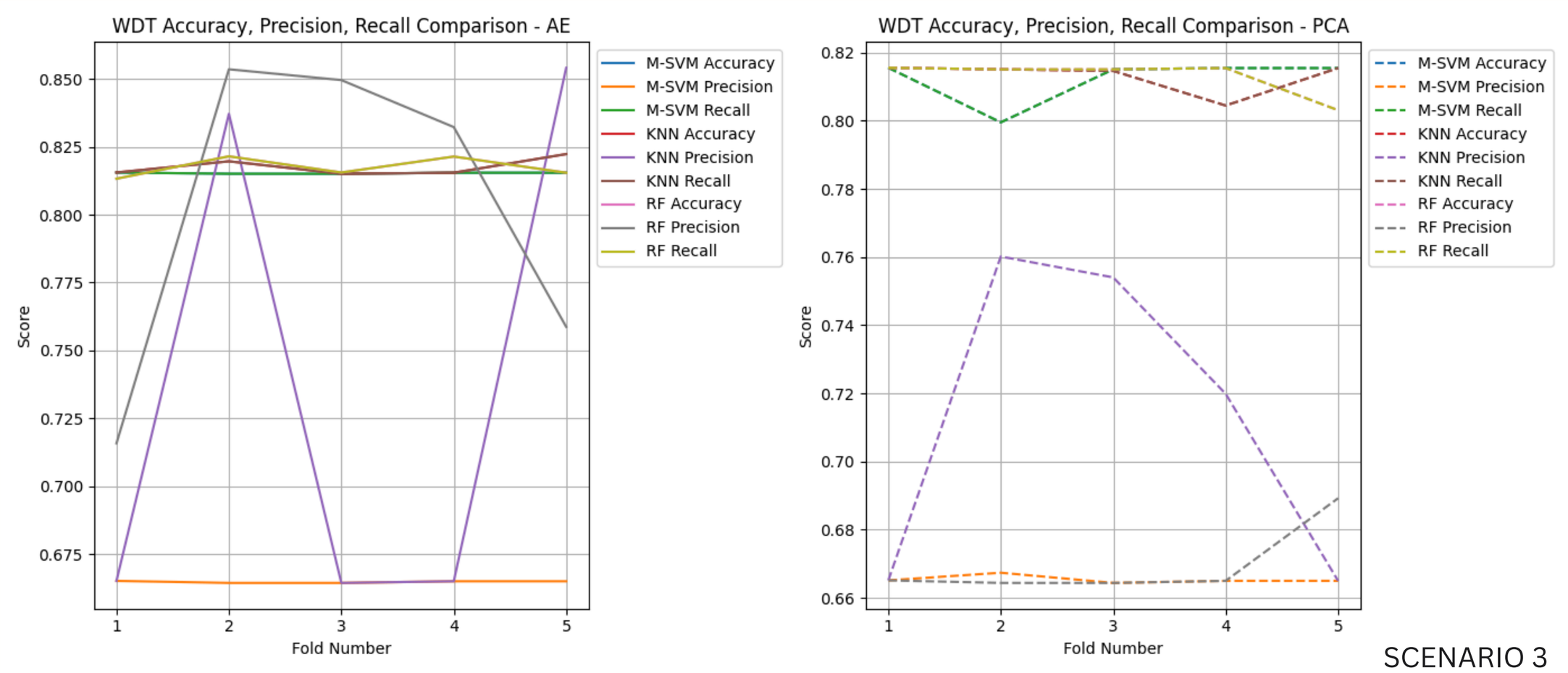


✅ **Scenario 1** → No exposure to attacks during training.
✅ **Scenario 2** → Tests generalization to an unseen attack.
✅ **Scenario 3** → Evaluates how models perform with minimal attack presence.

SCENARIO 2



- The WDT-AE has the most precision score
- Autoencoders performed well in unsupervised learning scenarios but required fine-tuning.



SCENARIO 3

- RF has the most precision and recall

✅ **Scenario 1** → No exposure to attacks during training.
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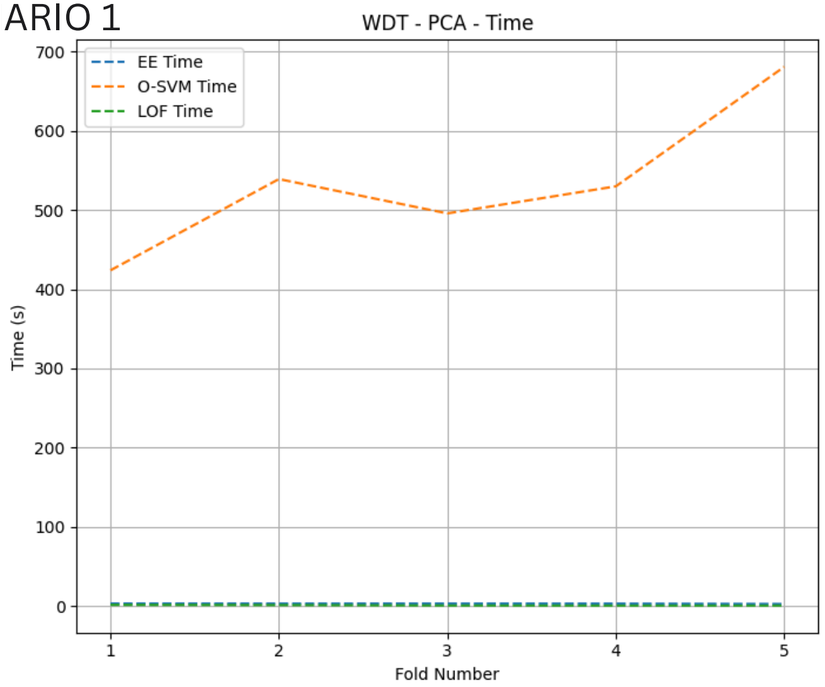
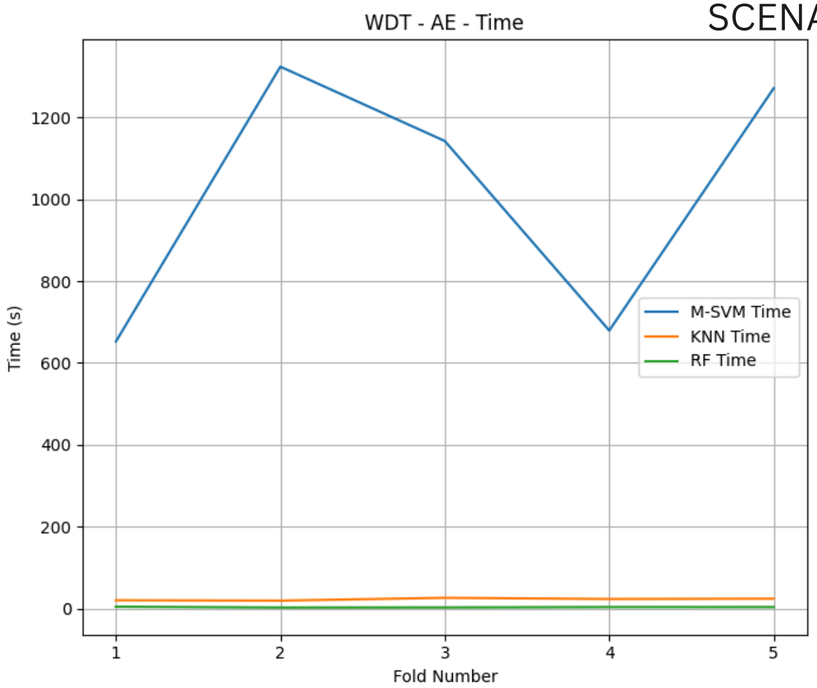
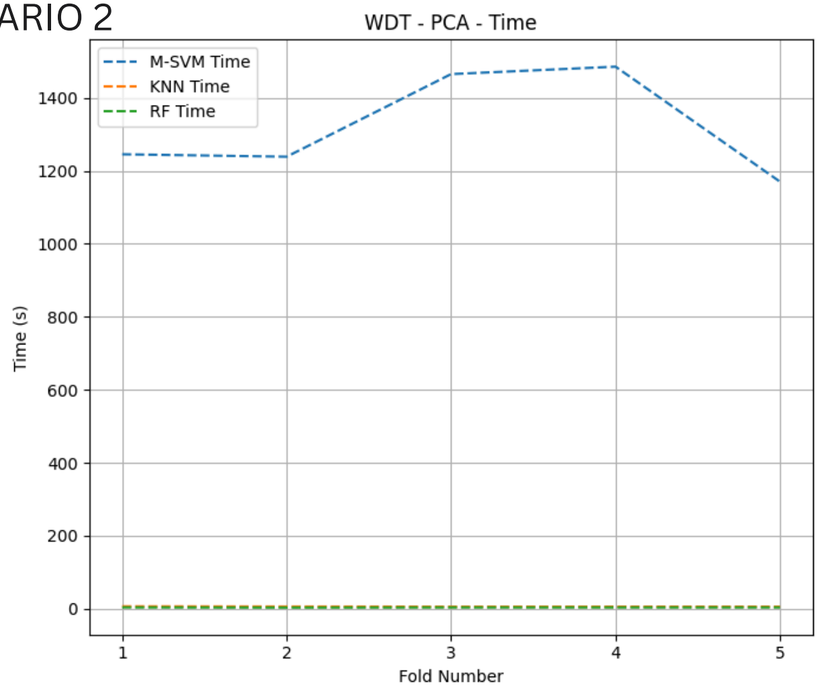
✅ **Scenario 3** → Evaluates how models perform with minimal attack presence.

SCENARIO 1



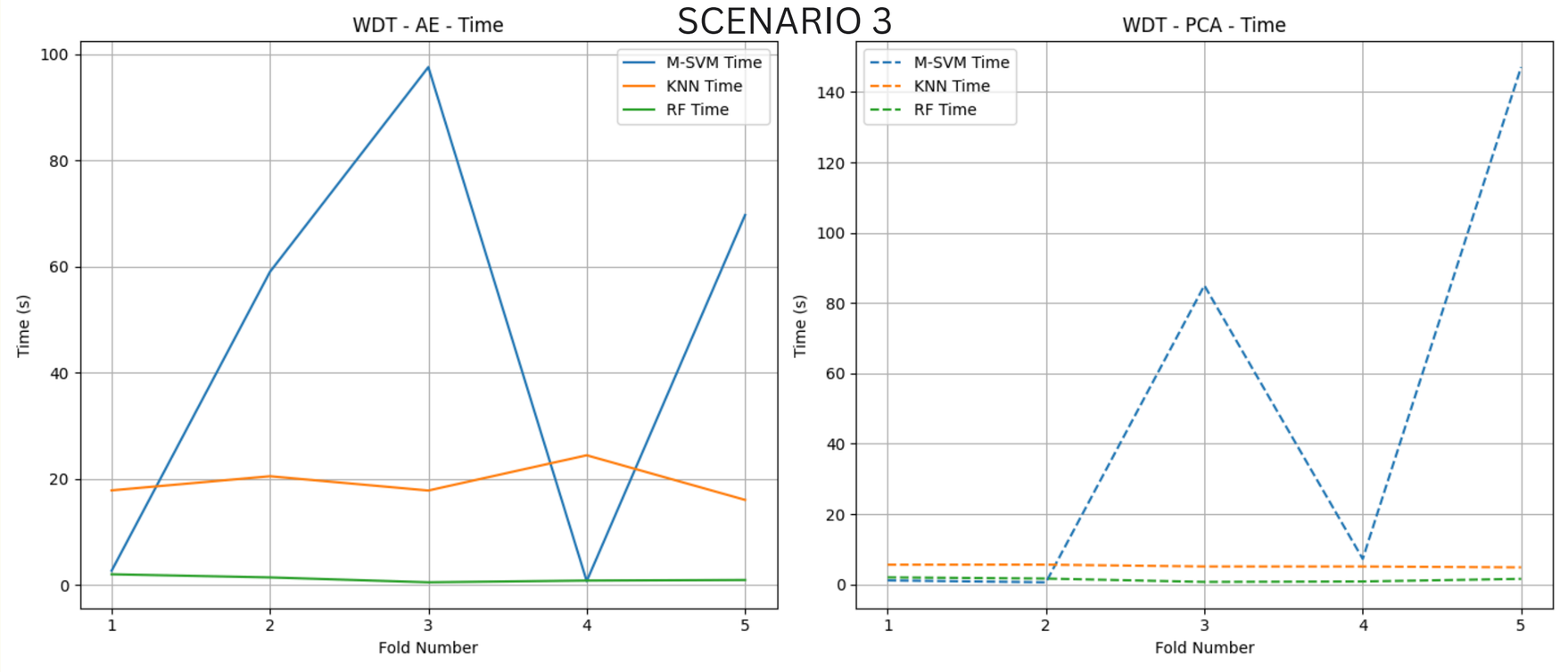
SCENARIO 2





- A plot comparing execution times for different feature extraction and classification methods.
- It helps identify the most efficient models for real-time ICS anomaly detection.

SCENARIO 3



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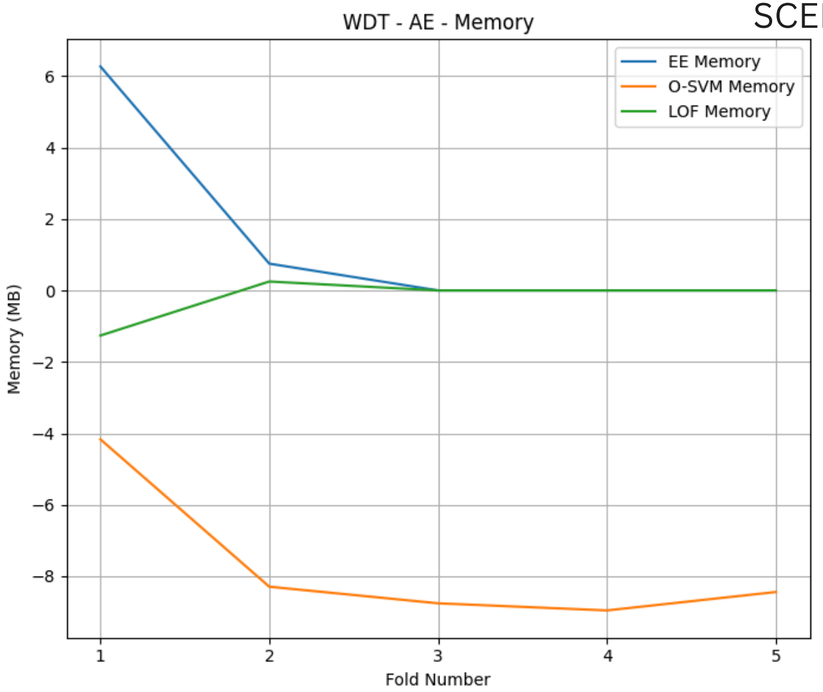

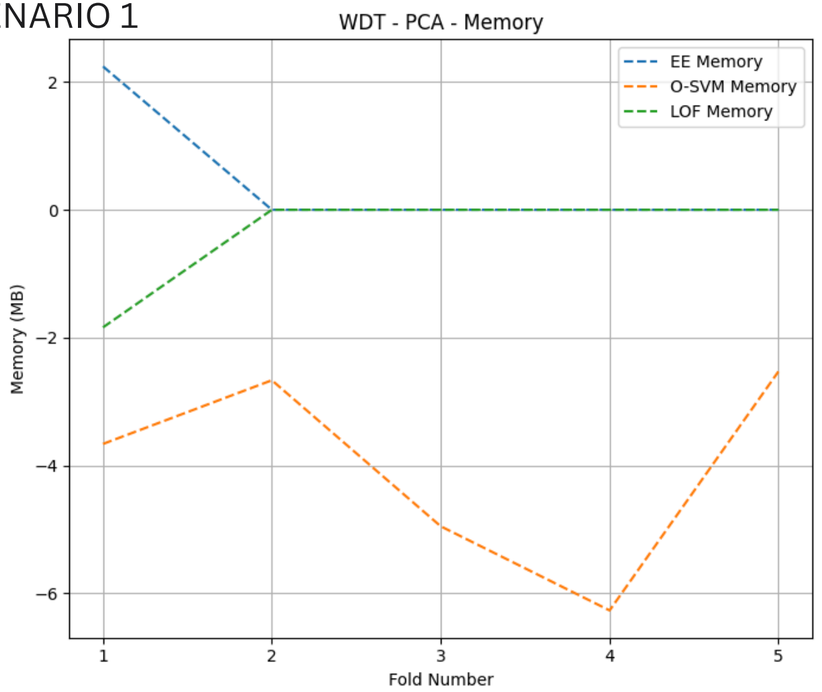

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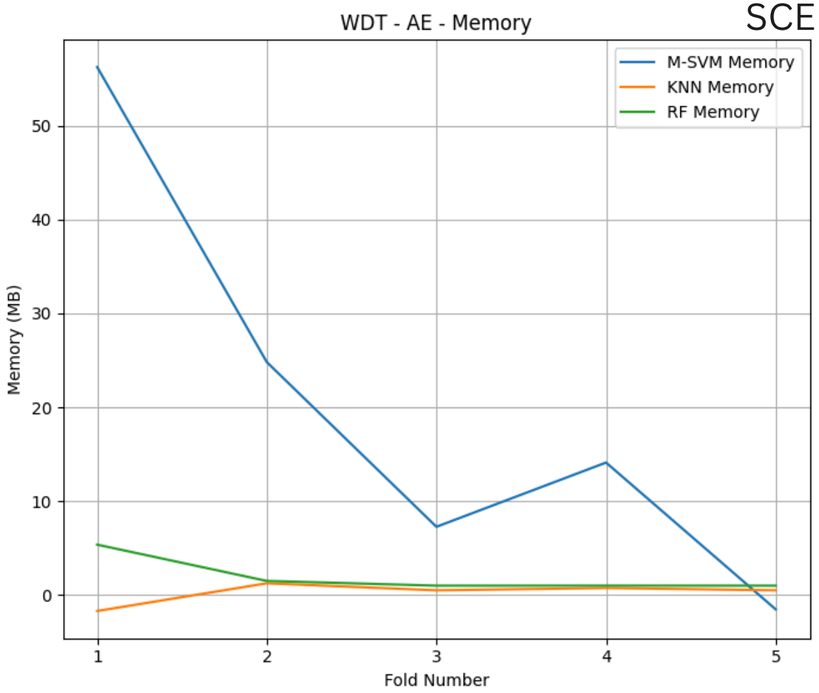
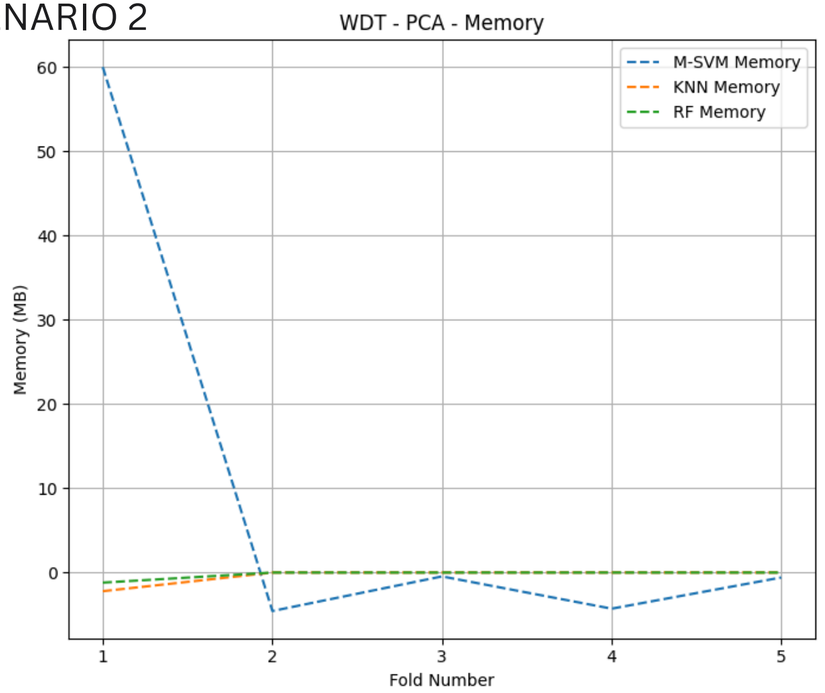
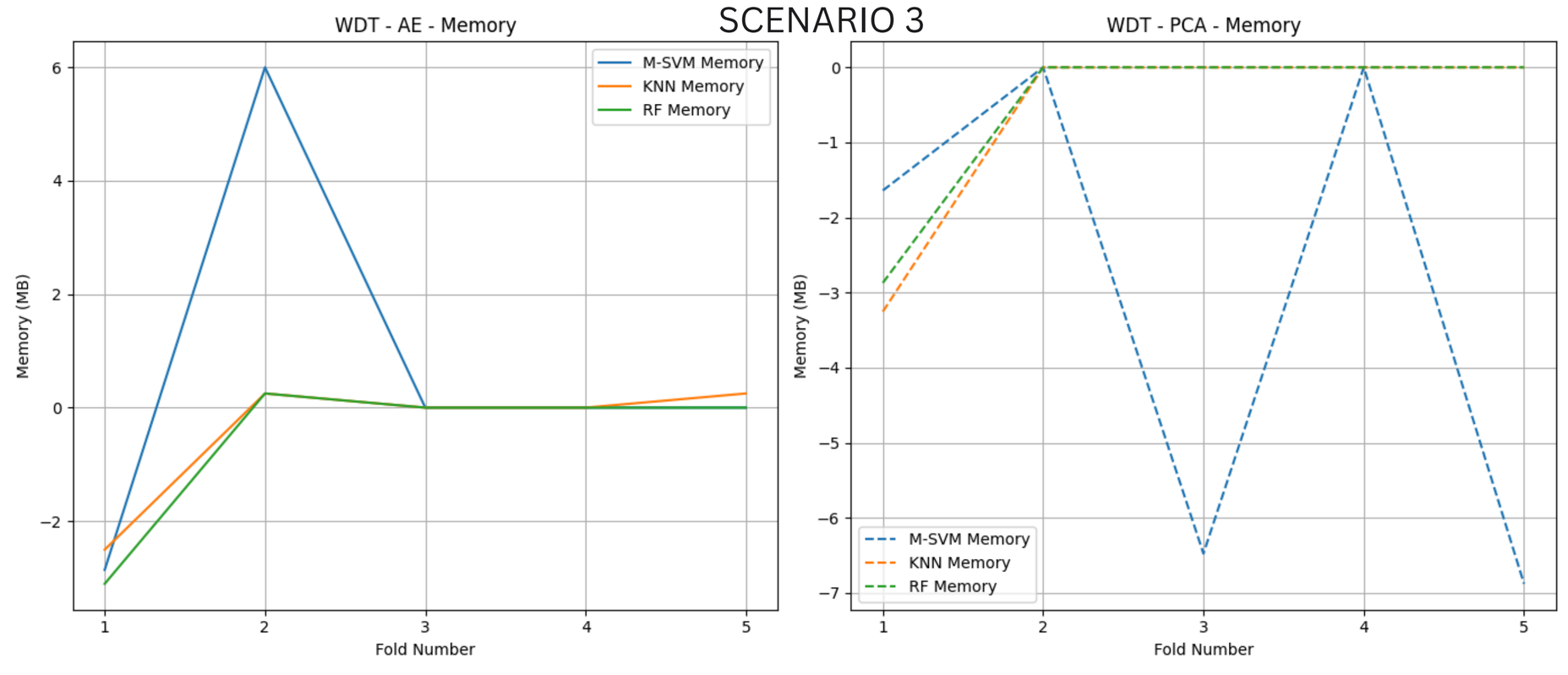




- A plot comparing memory consumption for different feature extraction and classification methods.
- It helps identify the most efficient models for real-time ICS anomaly detection.

SCENARIO 3



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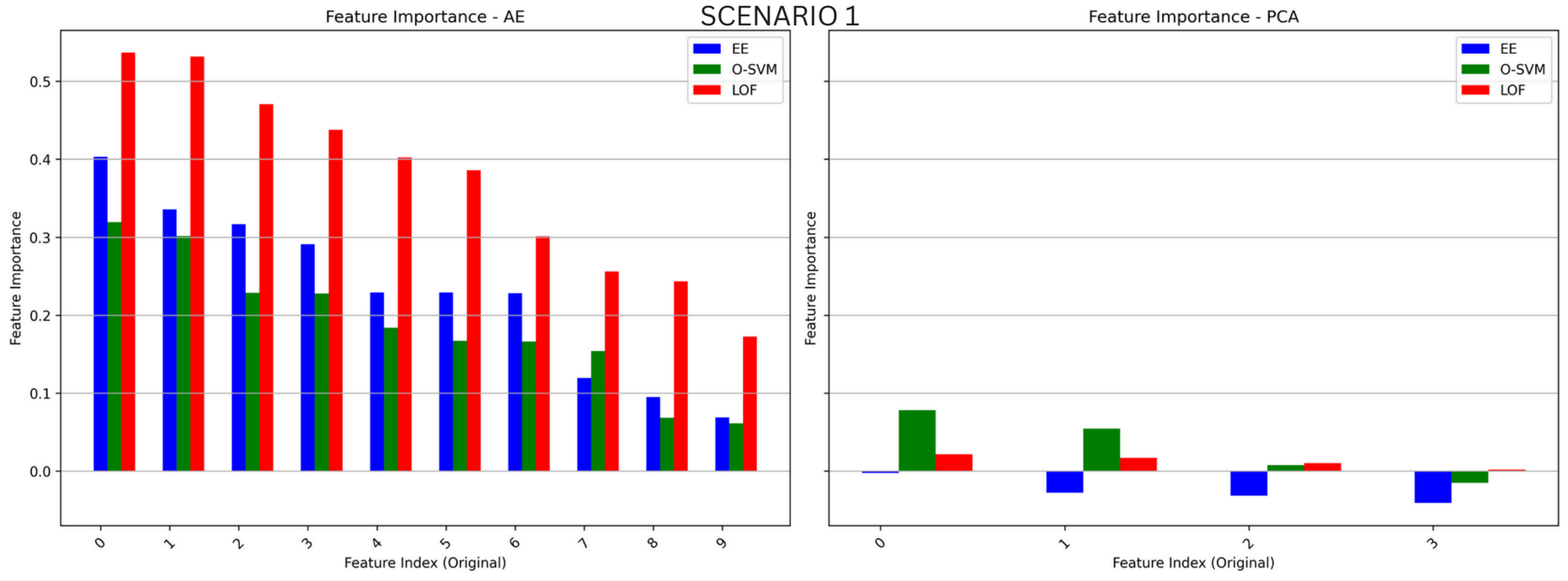

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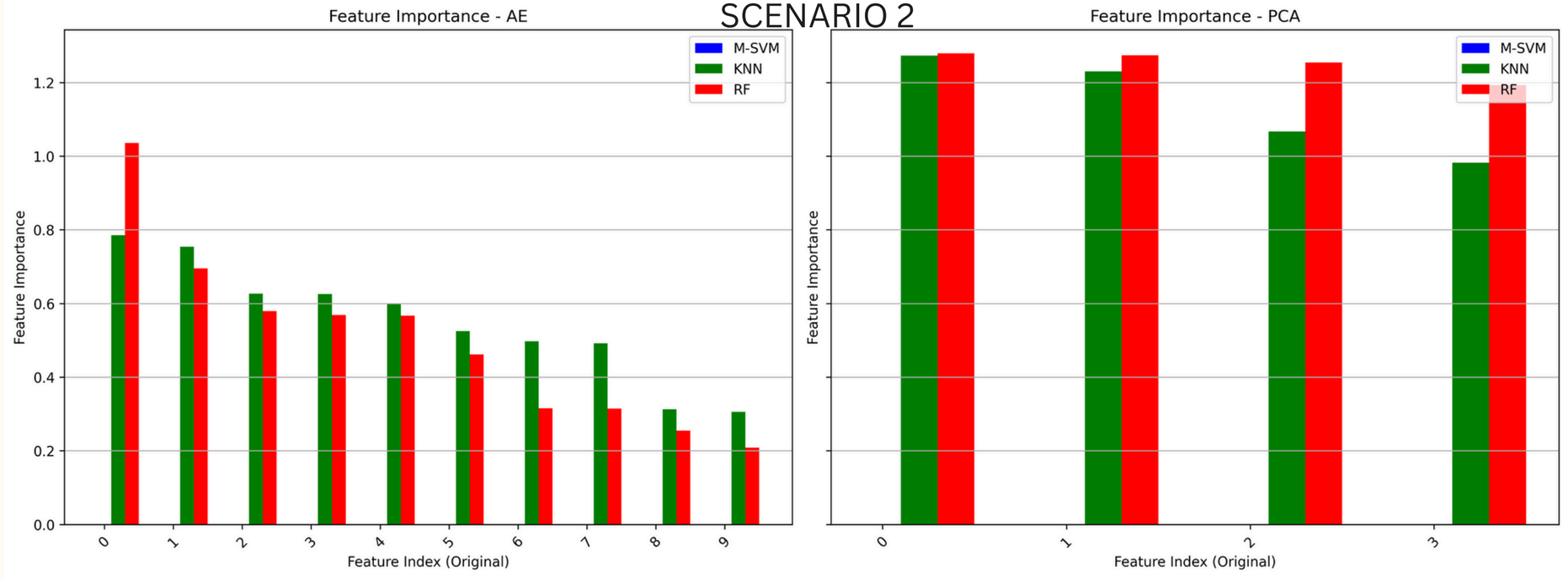
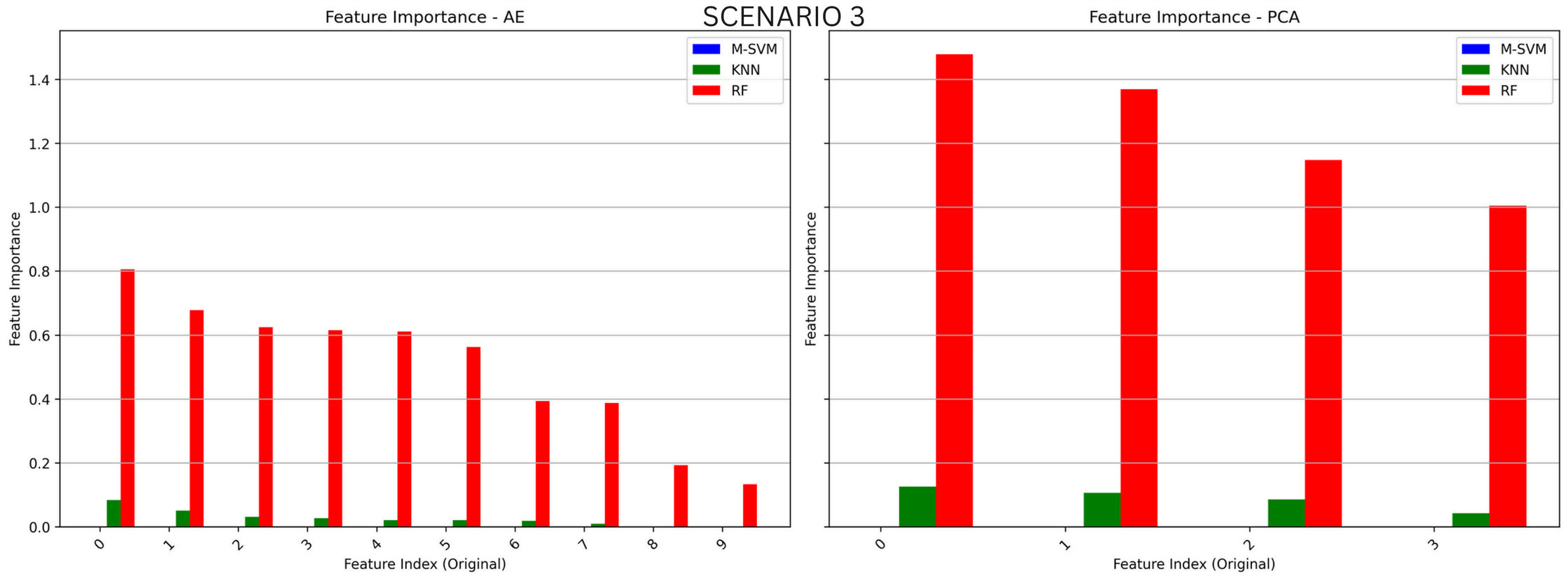


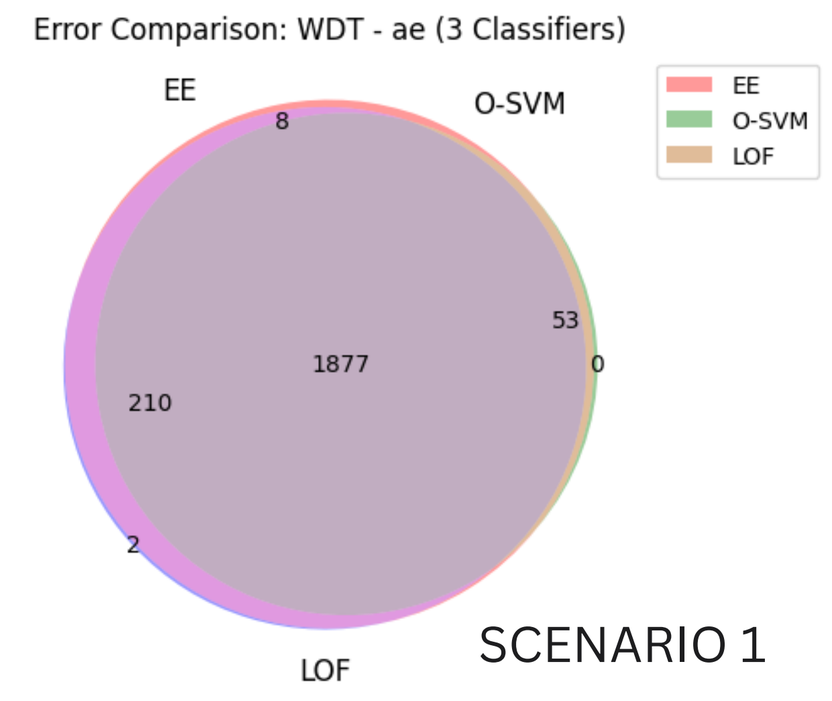
- A bar graph ranking the most influential features used in classification models.
- Identifying key features improves model interpretability and can help refine the dataset for better performance.

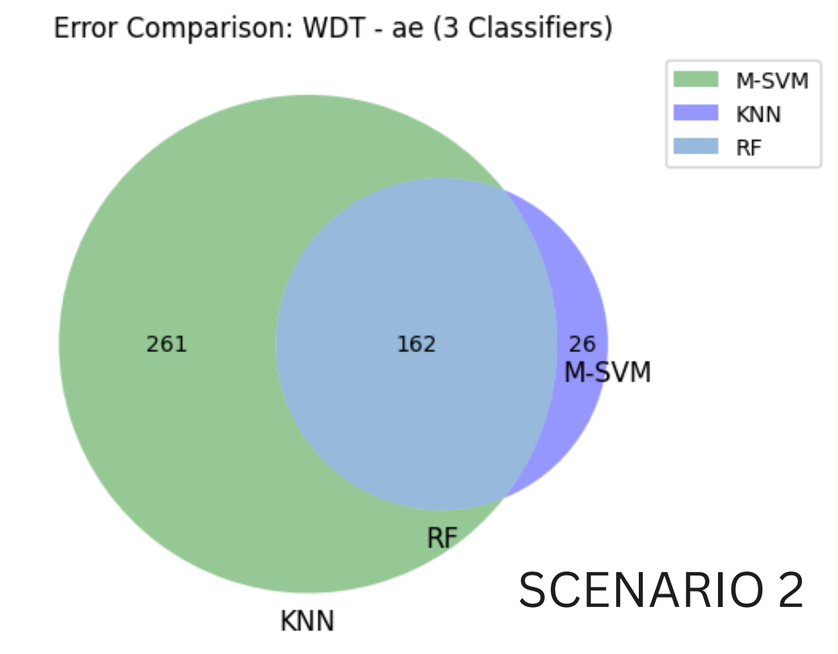
SCENARIO 3

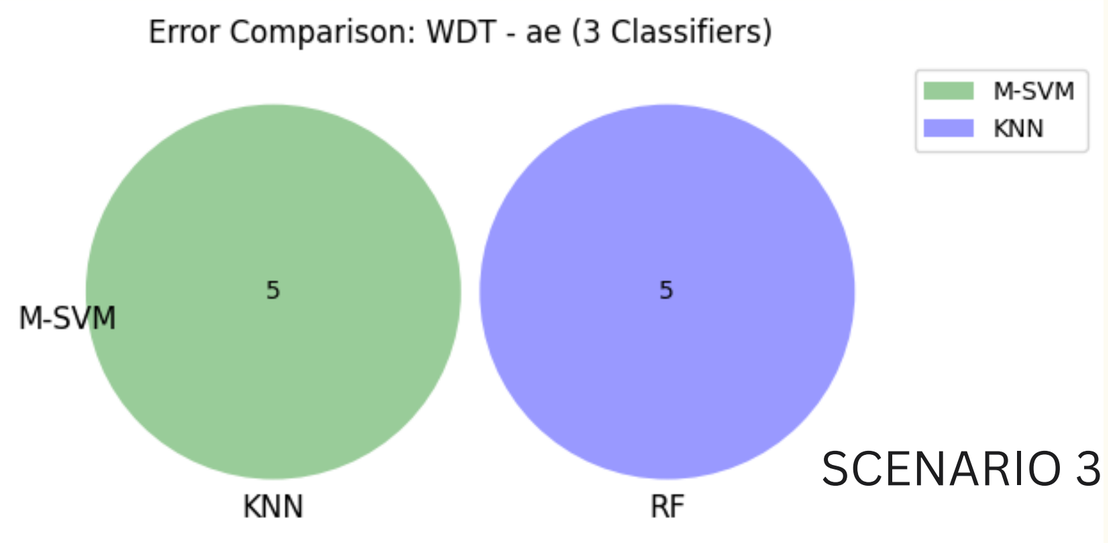


Feature importance plots for showing the most important features to the least important features

VENN - CLASSIFICATION ERRORS

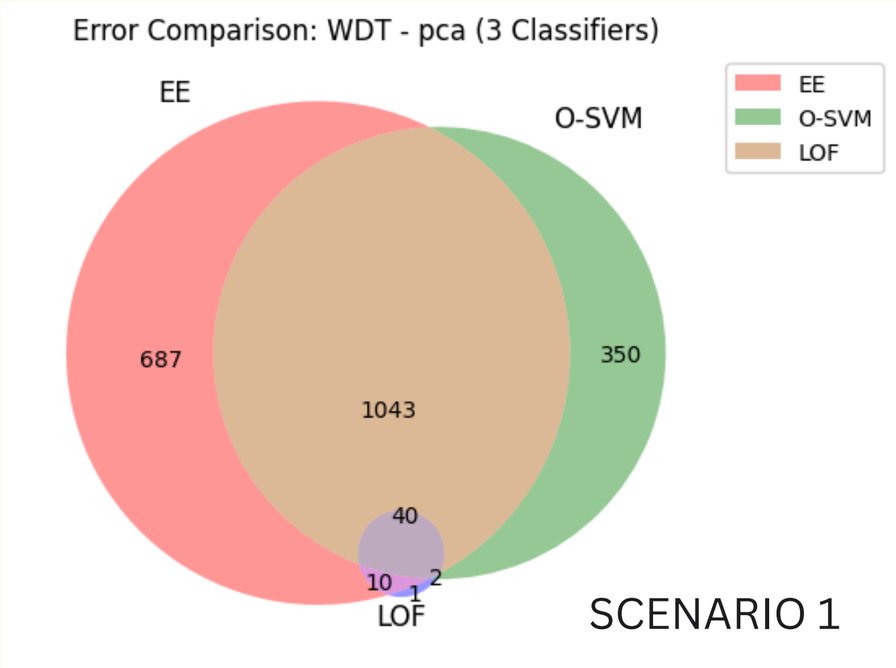


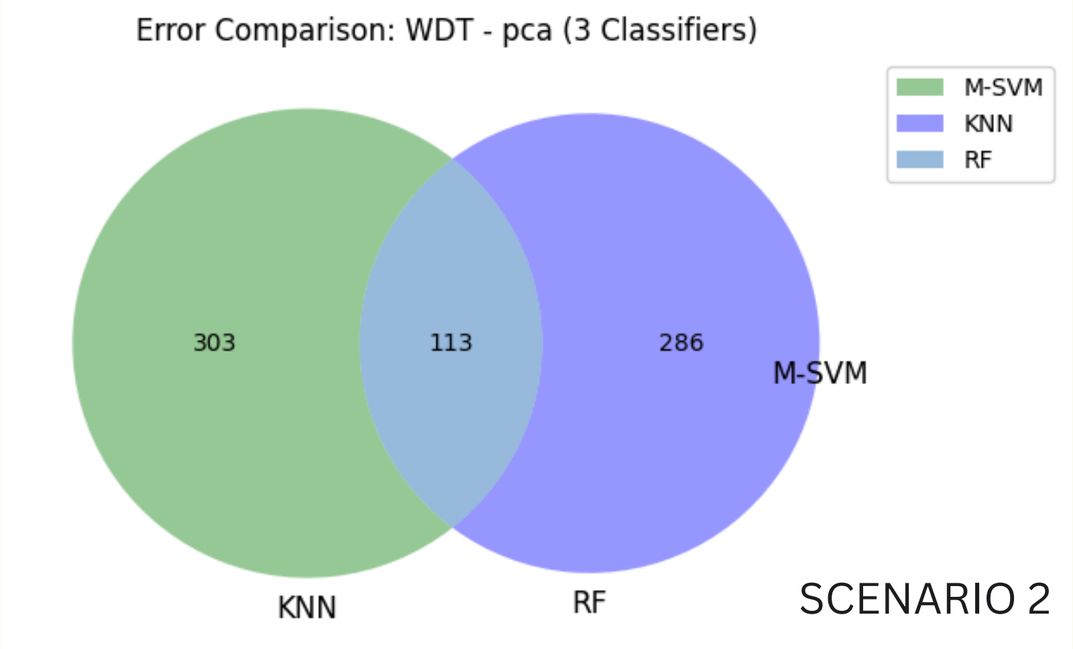




- A Venn diagram showing misclassification overlaps between different ML models.
- It helps analyze which errors are common across multiple classifiers, indicating potential dataset issues or model weaknesses.

VENN - CLASSIFICATION ERRORS





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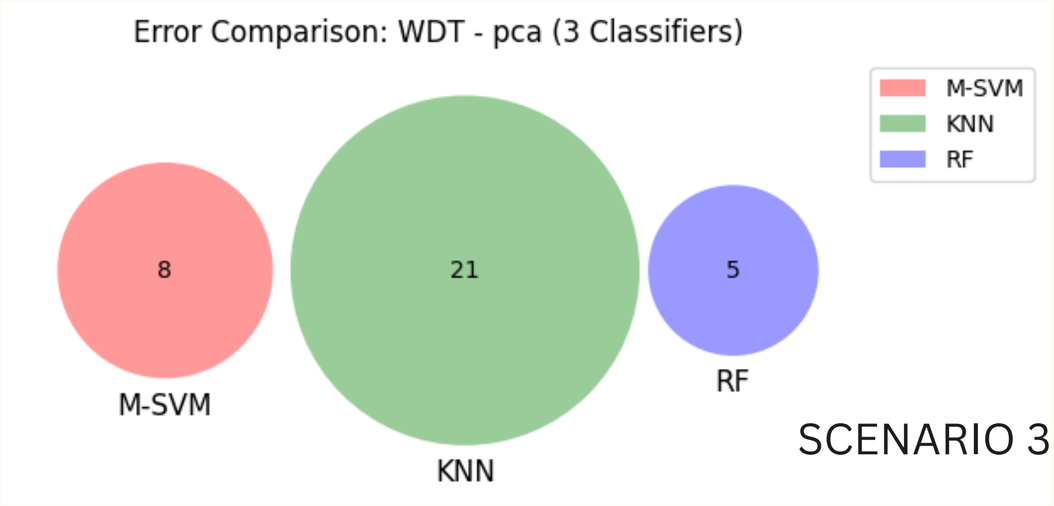


ENSEMBLE CLASSIFIER & DL-BASED CLASSIFIER

1️⃣ Deep Learning Classifier (CNN) – A Convolutional Neural Network (CNN) is trained on both sensor readings and network packets to detect anomalies.

2️⃣ Ensemble Classifier – Combines multiple ML models to improve detection performance using three voting strategies:

- Random Voting → Picks a prediction randomly from the ensemble models.
- Majority Voting → Takes the most common prediction among models.
- All Voting → Flags an anomaly only if all models agree.

ENSEMBLE CLASSIFIER & DL-BASED CLASSIFIER

Single-Class:

- *One-class SVM*
- *Local Outlier Factor (LOF)*
- *Elliptic Envelope*

Multi-Class:

- Random Forest
- k-Nearest Neighbors (k-NN)
- Multi-class SVM

WHY IS THIS IMPORTANT?

- CNNs can detect complex patterns that traditional models miss.
- Ensemble methods improve robustness by combining multiple models.

ENSEMBLE CLASSIFIER & DL-BASED CLASSIFIER

🔹 Scenario 1: Ensemble Classifier, Autoencoder as classifier and Statistical methods

📌 Objective: Evaluate ensemble classifiers and statistical methods on parsed physical readings and network packets.

📌 Evaluation:
✅ Ensemble Methods: Compare performance across different ensemble classifiers.
✅ Feature Sets: PCA (sensor data), Autoencoder (sensor data), Autoencoder (network data).
✅ Statistical Methods: Evaluate CUSUM & EWMA on parsed physical readings.
✅ Autoencoder as Classifier: Test on both sensor and network data.

📌 Key Question: Which ensemble classifier and statistical method provide the best precision and recall?

🔹 Scenario 2: Deep Learning Classifier Evaluation

📌 Objective: Compare deep learning-based classifiers on parsed physical readings and network packets.

📌 Evaluation:
✅ Train & Test Deep Learning Classifier separately for both data types.
✅ Optimize hyperparameters for best performance.

📌 Key Question: How effective are deep learning models in detecting cyber-physical anomalies?

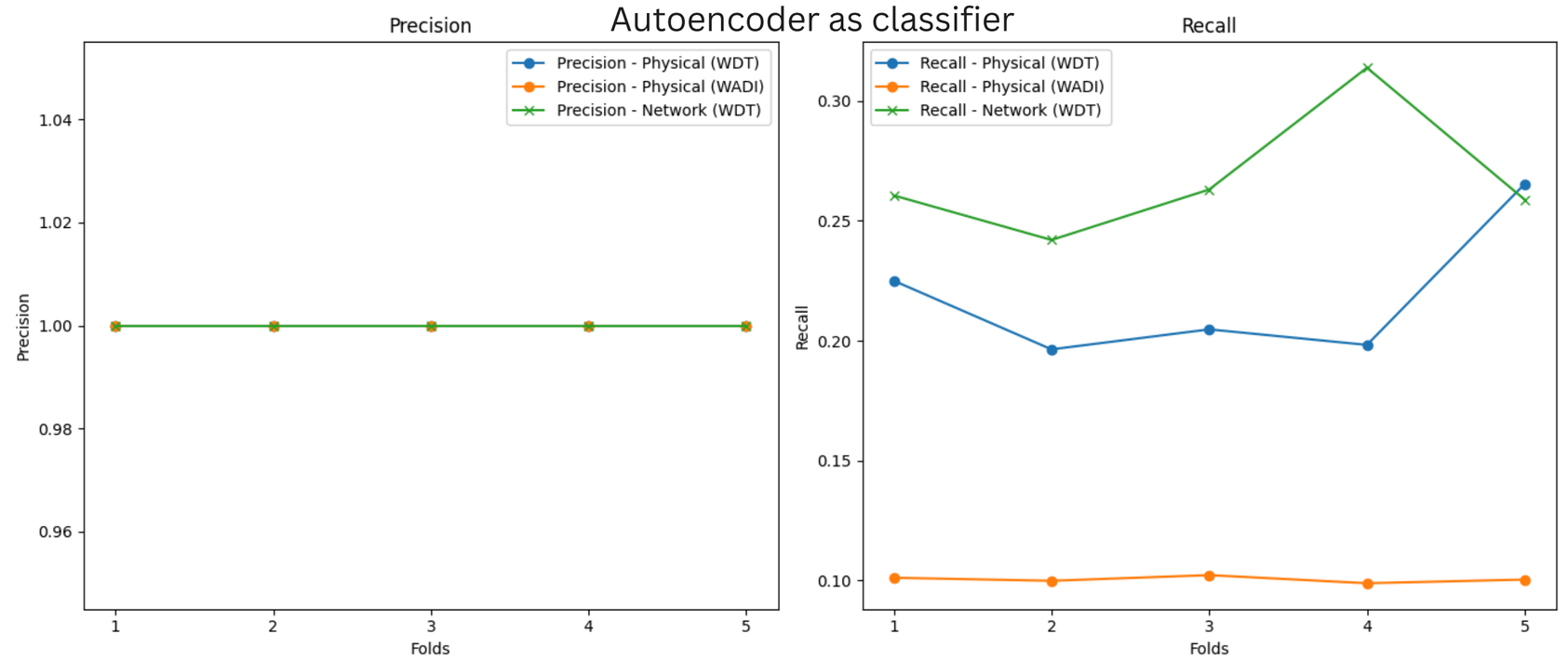
🔹 Scenario 3: Deep Learning Classifier in Different Conditions

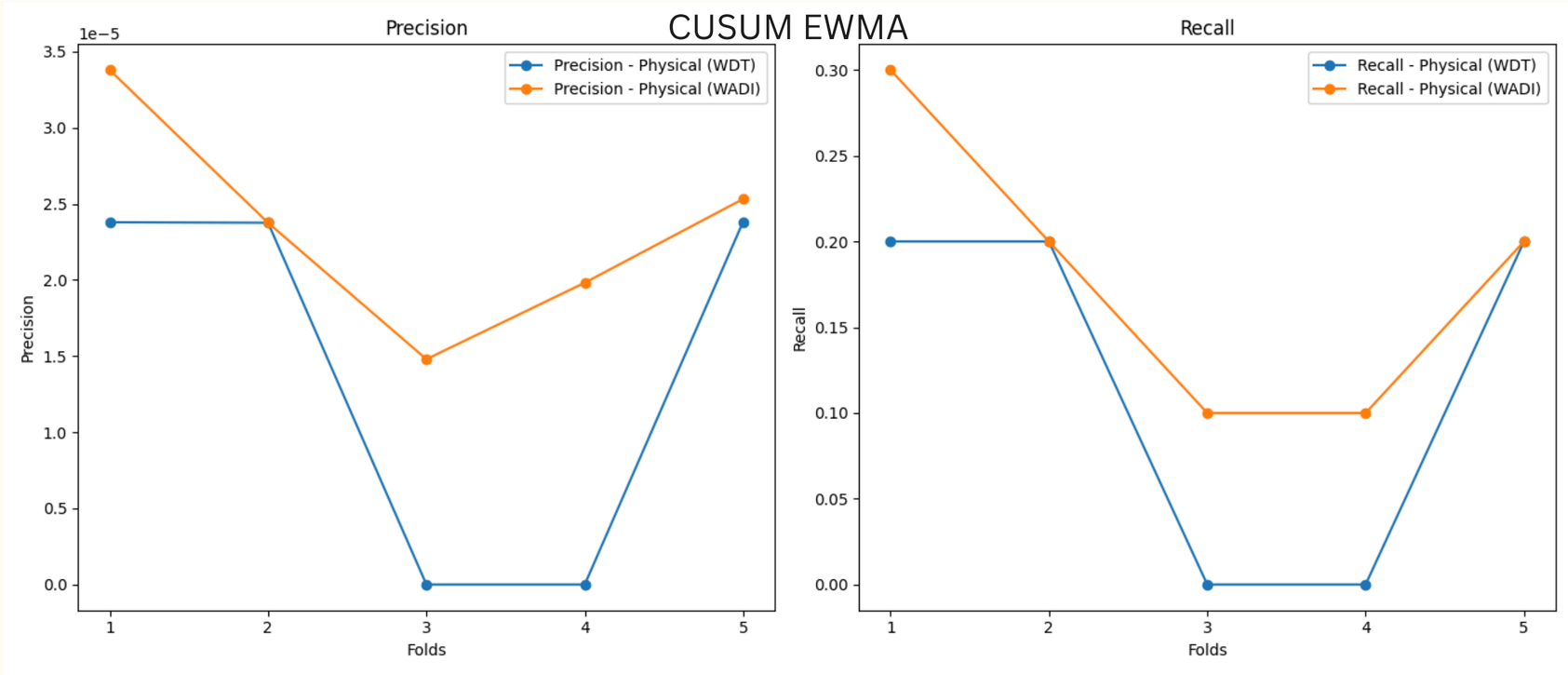
📌 Objective: Assess deep learning classifier performance under new evaluation conditions.

📌 Evaluation:
✅ Train & Test Deep Learning Classifier separately for sensor and network data under new conditions.
✅ Optimize hyperparameters to ensure high precision and recall.

📌 Key Question: Can deep learning classifiers adapt to different attack scenarios effectively?

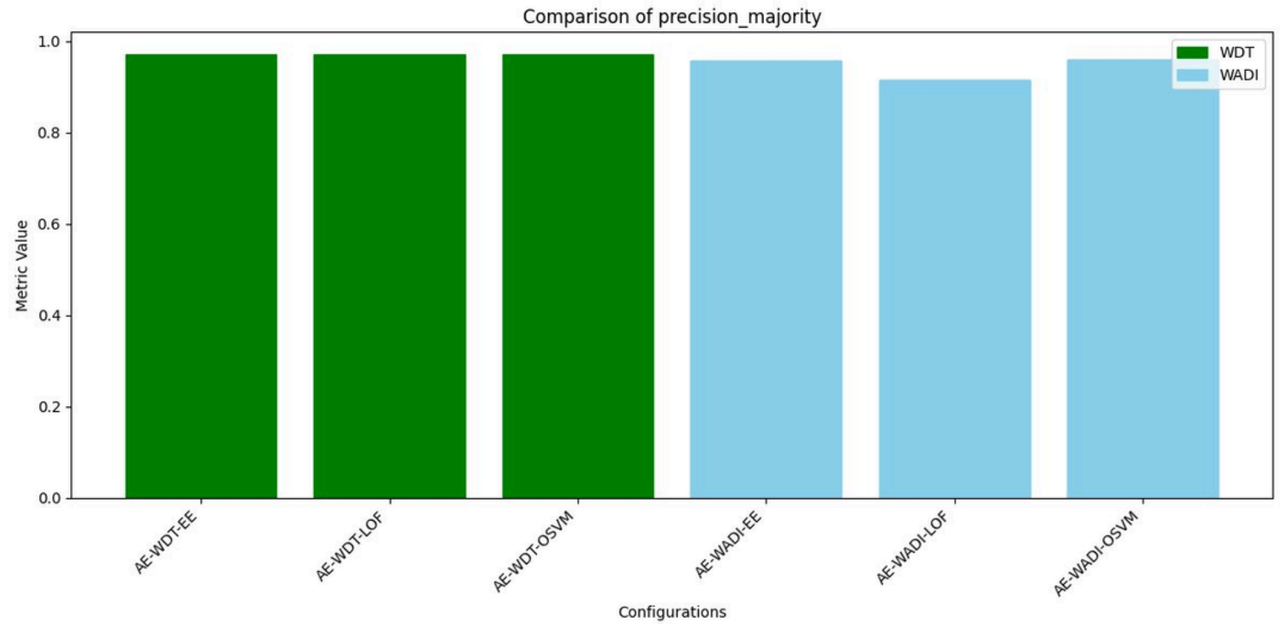
RESULTS

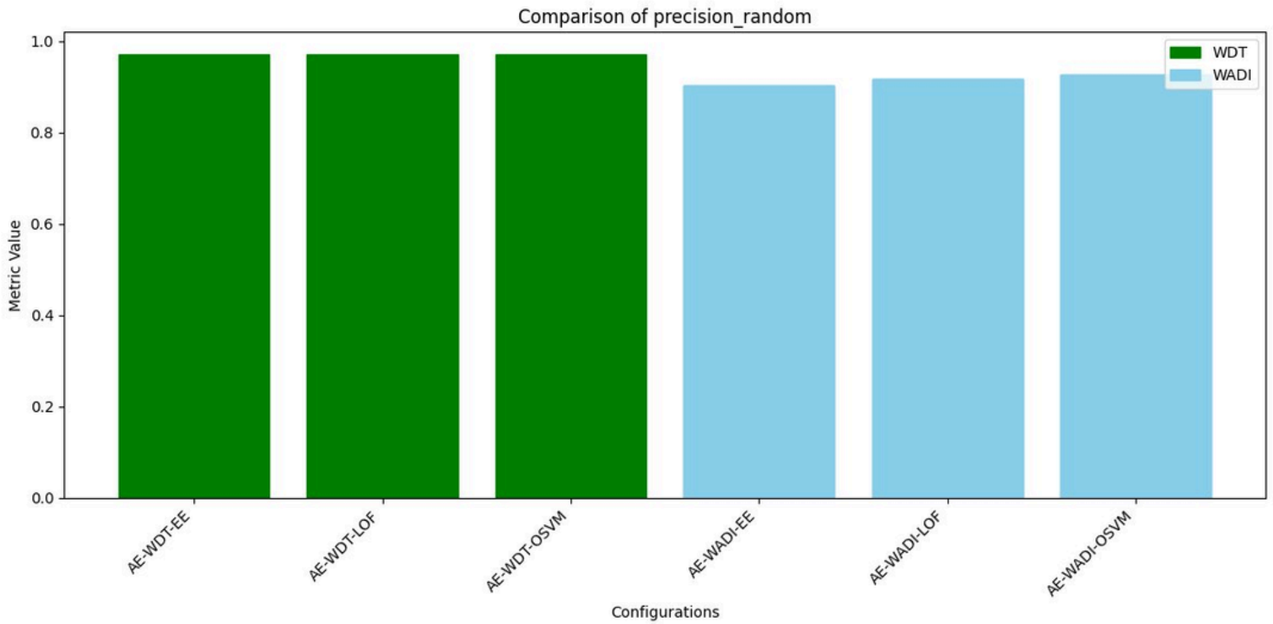


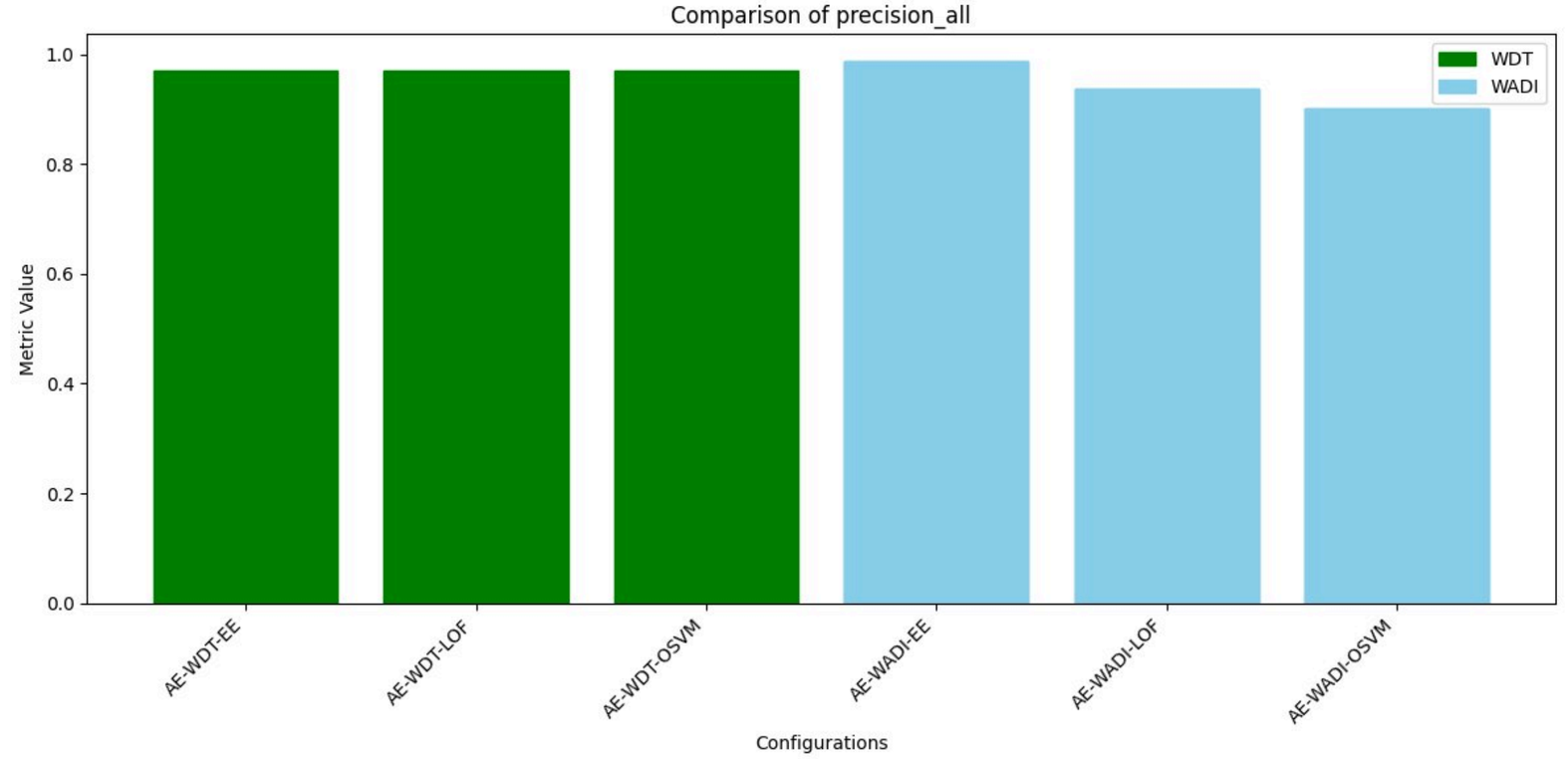


SCENARIO 1

- A line graph comparing precision, recall for different classification models.
- It helps evaluate the model's effectiveness in detecting true anomalies while minimizing false positives and false negatives.
- A higher precision indicates fewer false positives, while a higher recall ensures more anomalies are detected.

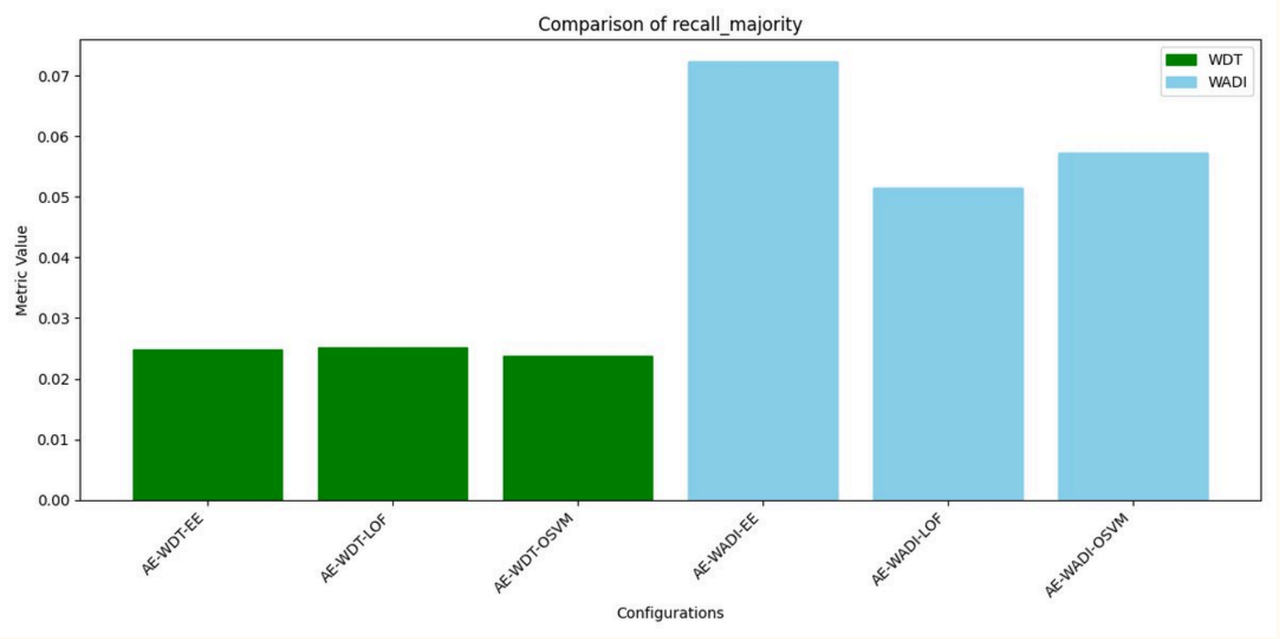


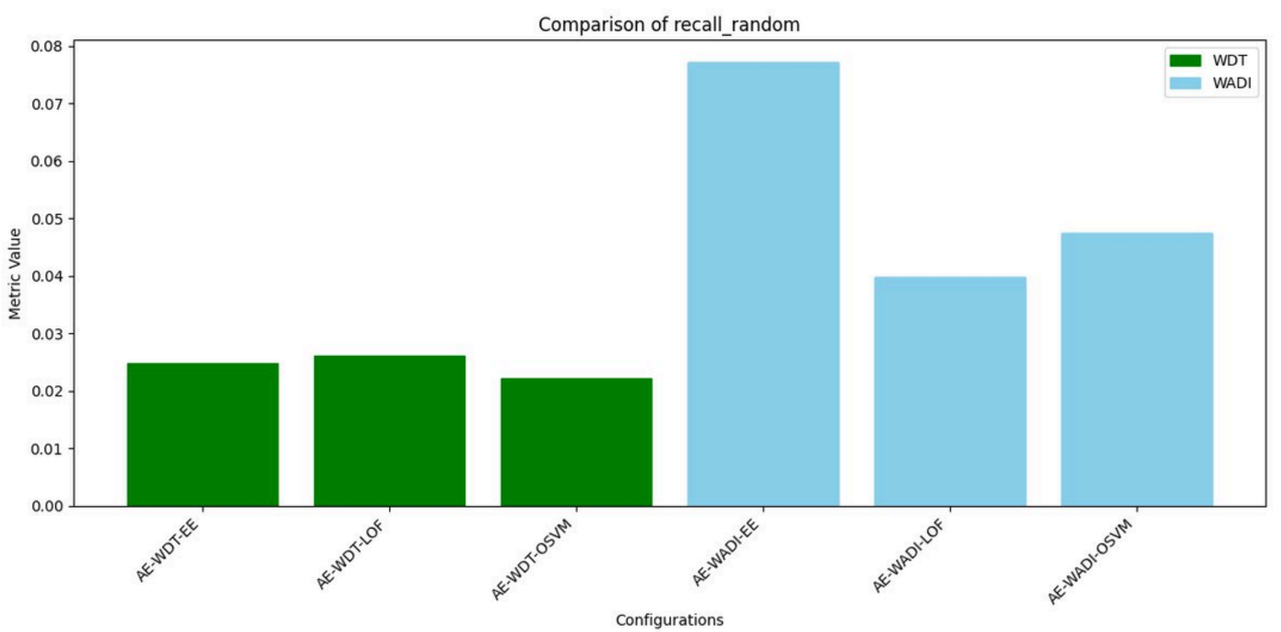


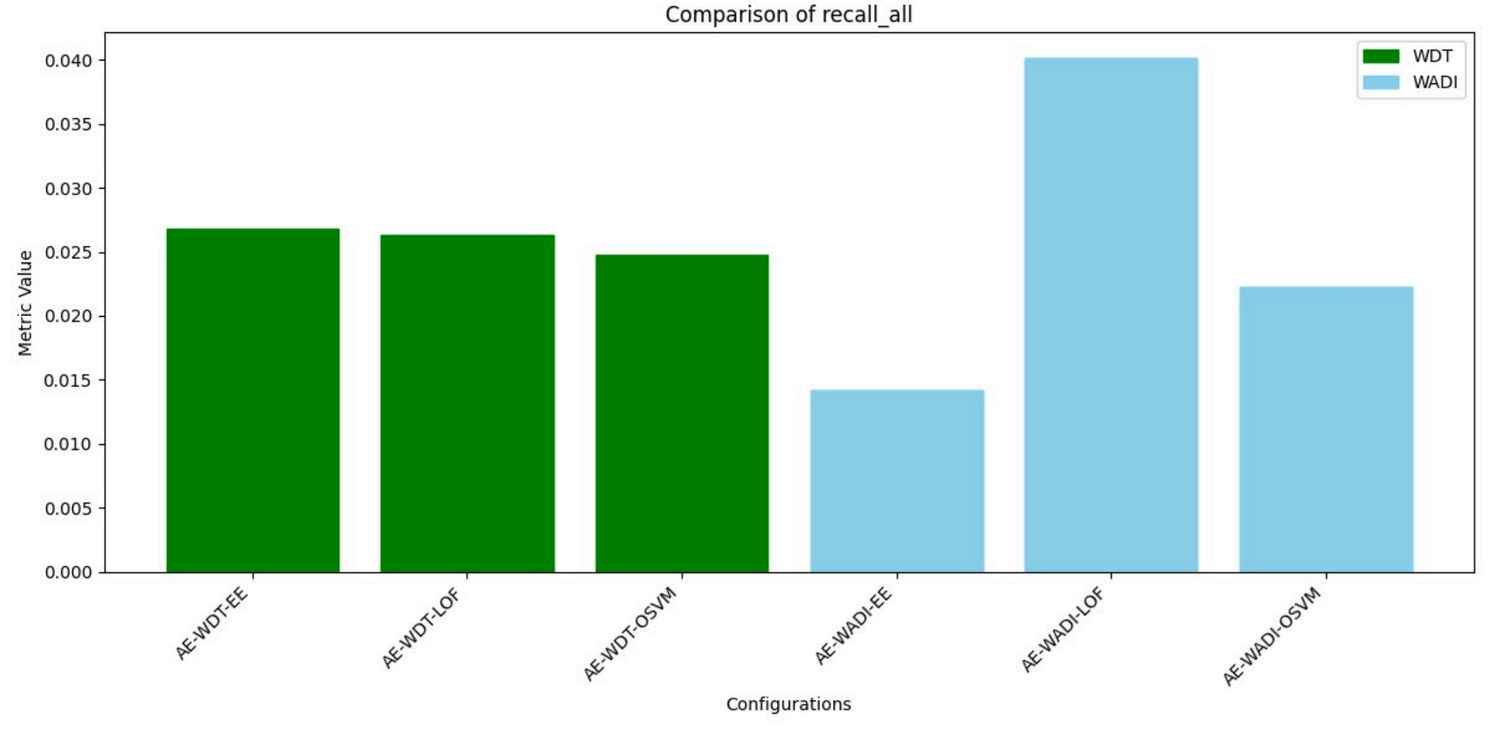


ENSEMBLE CLASSIFIER

- A bar chart comparing precision for different classification models.
- It helps evaluate the model's effectiveness in detecting true anomalies while minimizing false positives and false negatives.
- A higher precision indicates fewer false positives, while a higher recall ensures more anomalies are detected.



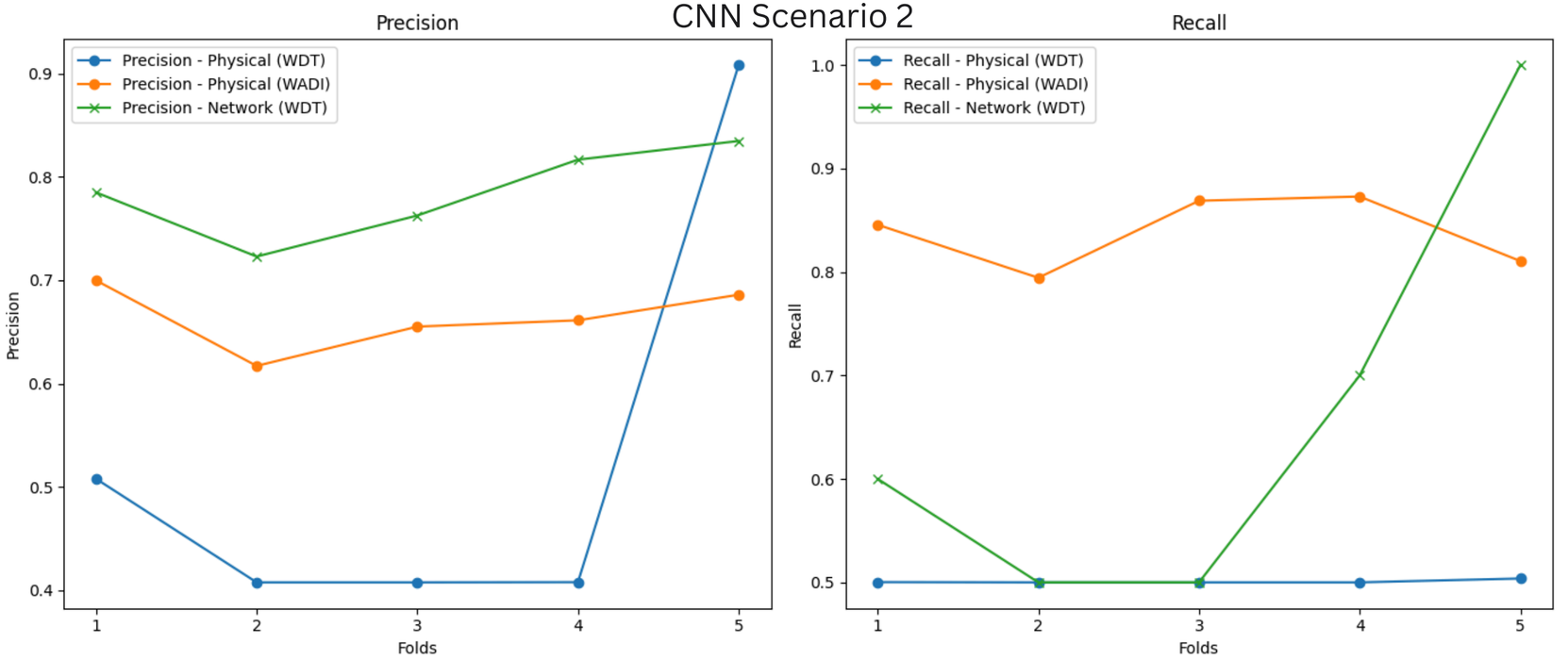




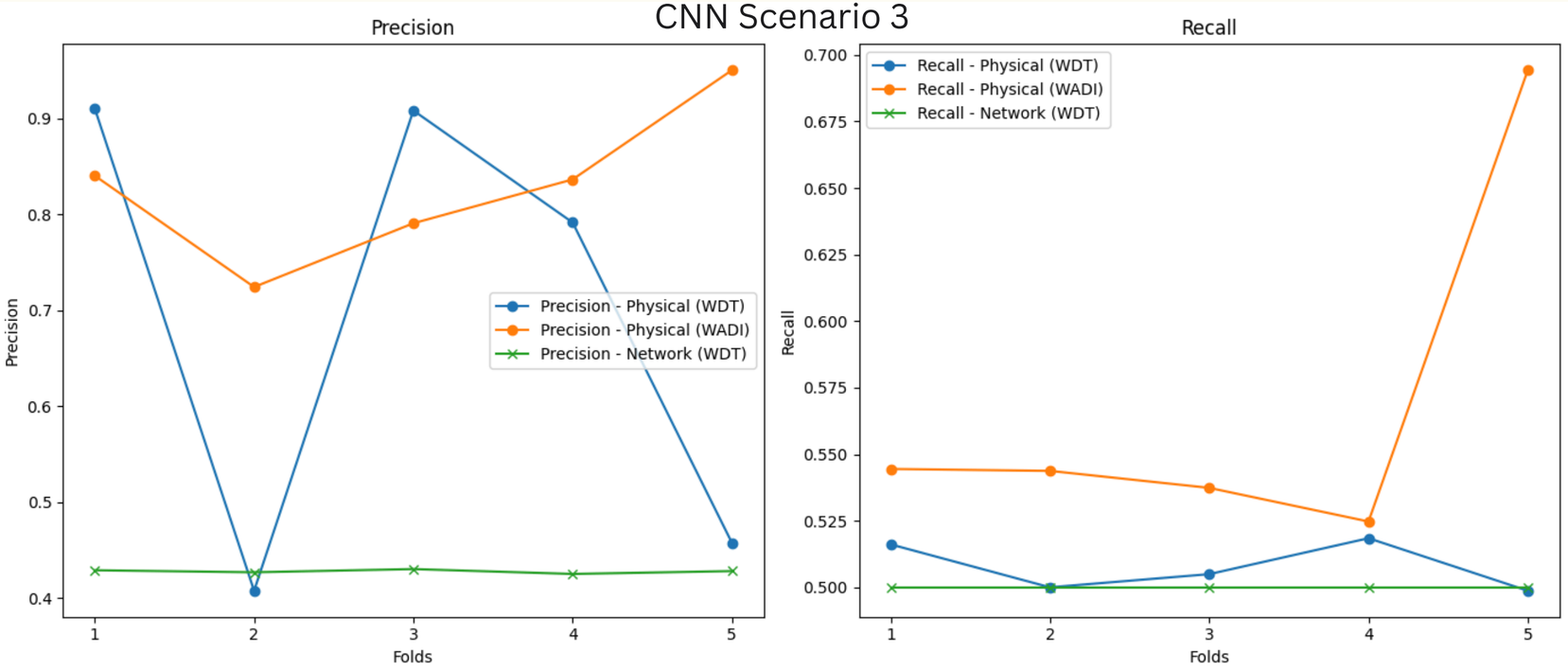
ENSEMBLE CLASSIFIER

- A bar chart comparing recall for different classification models.
- It helps evaluate the model's effectiveness in detecting true anomalies while minimizing false positives and false negatives.
- A higher precision indicates fewer false positives, while a higher recall ensures more anomalies are detected.

CNN Scenario 2



CNN Scenario 3



- A line graph comparing precision, recall for different classification models.
- It helps evaluate the model's effectiveness in detecting true anomalies while minimizing false positives and false negatives. A higher precision indicates fewer false positives, while a higher recall ensures more anomalies are detected.

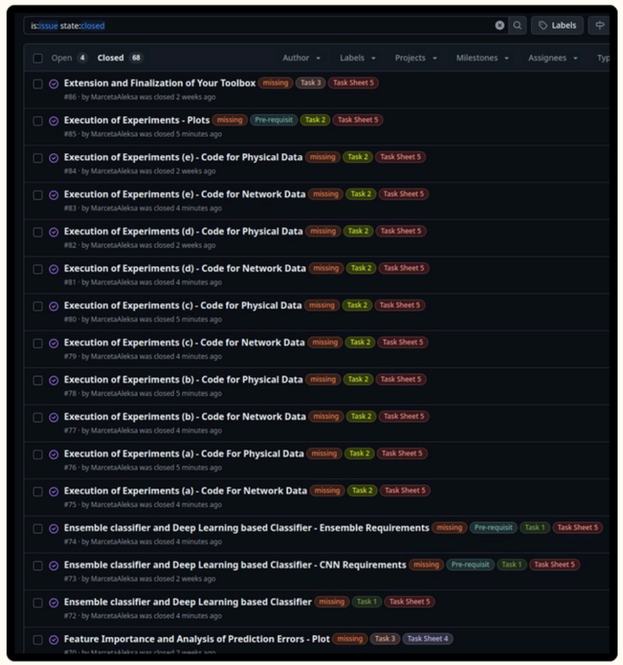
LESSONS LEARNT

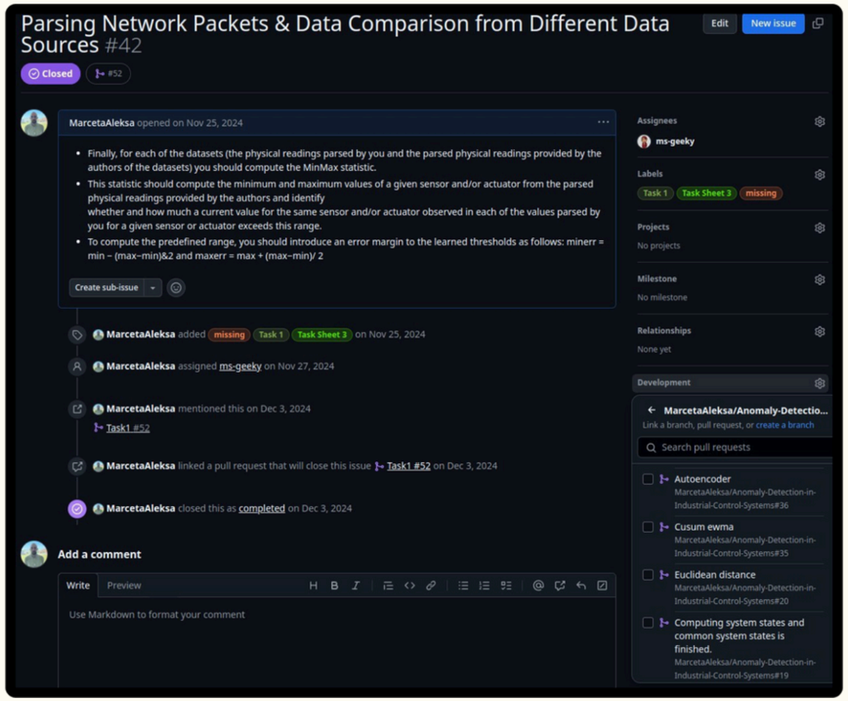
- Statistical methods provided quick but less flexible detection mechanisms.
- Traditional Machine learning models offered a balance between efficiency and detection accuracy.
- Deep learning models excelled in high-accuracy detection but required significant computational power.
- Ensemble learning improved overall robustness by combining strengths of multiple classifiers.
- Feature selection (PCA, Autoencoder, N-Grams) played a crucial role in optimizing model performance.

LIMITATIONS

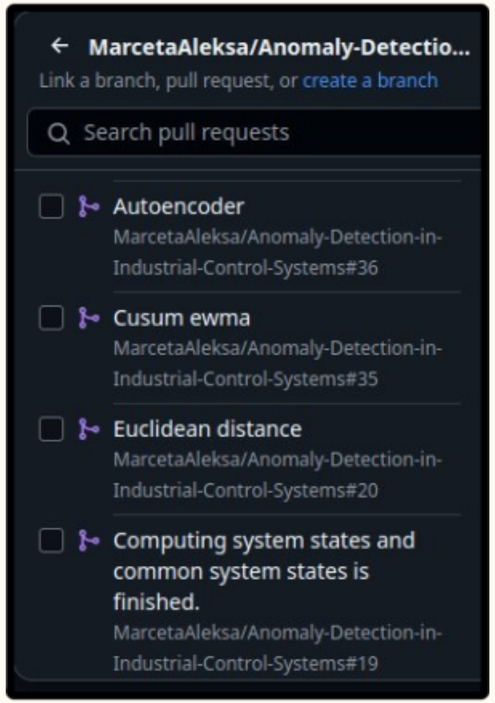
- Time constraints towards the end of the project led us to be more stressed resulting in slightly bad results
- Size of wadi dataset - large -> execution of scenarios took too long
- Description of attacks in datasets limited our approach because we had to look for patterns in attacks
- New approach - we looked for sequence of attacks present and one sequence would reflect one attack type
- Repetition of readings in different folds - inaccurate predictions because not representative of real world data
- In very few cases, we used manual tuning instead of automatically tuning parameters.

GITHUB CODE ORGANIZATION





- 💡 Collaboration Made Easy
- 💡 Task Management & Issue Tracking



THANK YOU

ANY QUESTIONS?